

Experiments with Theta Cycles Modulo p^2 : Hasse-Jet Duality and a Shifted-Conic Conjecture

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Abstract

Some weights in theta cycles modulo p^2 are unknown. We study one such region. We prove that its weight-lowering obstruction is a first Hasse jet. We also prove a perfect pairing for the Hasse quotients and show that theta is skew-adjoint for this pairing. Our minimal-weight projection and filtration arguments separate the boundary directions. Exact computations classify 692 ordinary cross-wedge cells. One shifted quadratic matches all 458 interior obstructions. The other 234 cells lie on three boundary lines. Two independent tests give 2,512 matches with one factorial boundary scalar. On the two triangular lines, we prove the unit and finite-block terms and compute the low-point jets. The full shifted-conic law still needs one factorial block identity and two triangular containments. These three identities remain conjectural. We provide exact code, negative controls, and tests that can find a failure.

Keywords: theta cycles; modular forms modulo prime powers; experimental mathematics; Hasse invariant; Bockstein class; weight filtration.

2020 Mathematics Subject Classification: 11F33, 11F11, 11Y40.

1 Introduction

The theta cycle of a modular form modulo a prime is classical. A new first-order obstruction appears modulo p^2 . A form may lower in weight modulo p but fail to lower in the same way modulo p^2 . Ahlgren, Raum, and Richter determined the initial segment and gave later exact values and bounds [2]. Some low points in the next band remain unknown for general primes.

The basic filtration theory modulo p comes from work of Swinnerton-Dyer and Serre [11, 10]. Katz gave a geometric theory of modular forms in characteristic p , including the Hasse invariant and the theta operator [8]. Later work on modular weights and companion forms developed this viewpoint further [6, 5].

We ask one question. Does the quadratic pattern in the resolved wedge remain valid after reflection into the next wedge? Our first candidate failed on every tested $D = 0$ cell. We then added the missing minimal-weight projection. In the interior, the lower Bockstein class followed

$$i'^2 + (k - 1)i' - (n + 1)(n + 2) \quad \text{in } \mathbb{F}_p. \quad (1)$$

Every disagreement occurred on one of three affine lines. The full residuals suggested one factorial block and two triangular endpoint formulas.

We explain how we found and tested this formula. Every calculation uses exact modular arithmetic and Sturm-safe remainders. The tests include a failed baseline, a blind extension to new primes, and a separate calculation in a two-dimensional cusp-form space. We then prove the geometry behind the experiment. The weight-lowering obstruction is a first Hasse jet. The ambient quotient has a perfect pairing, and theta is skew-adjoint for this pairing. Multiplication by the integral Hasse inclusion commutes with theta modulo p^2 . We also construct the minimal-weight projection. Exact filtration separates the proposed boundary directions.

Three identities remain open. The direct formula should reduce to a factorial block identity. That identity would give the generic shifted-conic law and the observed scalar on the third line. Two triangular containments remain open on the other boundary lines. We compute their low-point jet terms for all admissible parameters. We state the three remaining claims as conjectures because our computations cover only a finite range.

The open identities are explicit coefficient problems in known block bases. Their parameters satisfy short exact recurrences. Each identity also has a direct finite-field test. These facts suggest possible proof methods and make counterexamples easy to find. They do not replace a proof for all admissible primes.

Scope. We work with ordinary level-one forms and primes $p \geq 5$. The main census ends at $p = 43$; separate audits reach $p = 79$ and $p = 101$. We do not treat non-ordinary forms, higher level, or the three open block identities as settled cases.

1.1 The main idea in plain language

A modular form has a Fourier expansion, which is an infinite list of coefficients. The theta operator changes this list by multiplying the coefficient of q^n by n . A theta cycle records the smallest possible weight after each repeated use of theta. The pattern modulo p is well understood. Modulo p^2 , two forms can agree modulo p but differ at the next p -adic level. That first difference is the obstruction studied here.

Most tested positions follow one shifted quadratic rule. Three straight boundary lines need extra correction terms. The geometric results explain where the obstruction lives and why the boundary terms are independent. The exact computations then identify the formulas that a complete proof must establish. Three coefficient identities are still open, so the paper keeps the final shifted-conic law as a conjectural application.

The experiments identify the classification problem and the three boundary cases. The theorems explain why the candidate has this form. They also reduce the open part to explicit identities.

1.2 Main results and limits

The results fall into three groups.

1. **Uniform theorems.** We prove the first-Hasse-jet description, perfect Hasse-quotient duality, theta skew-adjointness, the explicit minimal-weight projection, separation of the boundary directions, exact affine recurrences for the factorial target, and exclusion of the finite block from both triangular endpoints. We also give a closed formula for the reset jet and prove that both low-point jet corrections lie below the endpoint cutoffs.
2. **Exact finite results.** The main census has 692 cells. Two later tests give 2,512 exact matches with the proposed third-line scalar. A separate index audit checks 210,176 parameter cells through $p = 101$.
3. **Open claims.** One factorial block identity and two triangular block-basis containments are still conjectural. The full shifted-conic law is conditional on these identities.

We use this split throughout the paper. A finite calculation is never stated as a theorem for all primes.

1.3 Relation to prior work

Ahlgren–Raum–Richter [2] prove the initial segment, further exact values or bounds modulo p^2 , and a quadratic condition for the exceptional positions in their resolved region. Our code also

checks the resolved $D = 2$ subwedge, whose conic is

$$i'^2 + (k - 1)i' - n(n + 1) = 0.$$

We study the complementary $D = 0$ subwedge. Reflection changes the constant term to $(n + 1)(n + 2)$. A new first-Hasse-jet projection is needed before this expression gives the actual obstruction. This is not a restatement of Ahlgren–Raum–Richter. The geometric pairing alone also does not prove the shifted-conic law.

1.4 Guide to the paper

Section 2 describes the finite experiment, the negative controls, the validation tests, and the failure rules. The next two sections give the first-Hasse-jet and direct block calculations. Section 6 separates the finite evidence from the open claims. The final sections state the conditional shifted-conic consequences, list the status of each theorem, and give exact commands for reproducing the computations.

2 Experimental design and discovery path

2.1 Objects, arithmetic, and inclusion rules

We study the filtration of $\theta^i f$ modulo p^2 for ordinary level-one modular forms of weight $4 \leq k < p - 1$. Every comparison uses exact integer arithmetic. At each relevant weight, we keep the Fourier expansion past the Sturm bound. Equality of the retained series therefore proves equality of the modular forms in that tested case. We use no floating-point approximation, numerical tolerance, random sample, or fitted model.

The computational record contains 5,572 cells. Of these, 5,434 satisfy the ordinary condition $k < p - 1$. The main experiment uses 692 ordinary, nondegenerate cells. For each such cell, a $D = 0$ index and its Tate complement lie in the proved $D = 2$ wedge. The cells cover primes $17 \leq p \leq 43$ and weights $8 \leq k \leq 22$. We select a cell from these arithmetic conditions alone. We do not inspect its obstruction class or the candidate quadratic before selection.

For each cell, the program performs four exact operations:

1. reduce $\theta^i f$ modulo p^2 against an integral echelon basis of M_{w-} ;
2. divide the resulting canonical remainder by p and reduce modulo p , producing the lower Bockstein class $\mathcal{B}_i$;
3. independently construct the shifted leading-block candidate

$$\mathcal{L}_i = q_1(n, i') X^p \theta^{i-p} f, \quad q_1(n, i') = i'^2 + (k - 1)i' - (n + 1)(n + 2);$$

4. compare the complete canonical remainders, not a selected coordinate or a short Fourier prefix.

The program records both agreements and failures. For each failure, it stores the full residual $\mathcal{B}_i - \mathcal{L}_i$ whenever the proposed identity fails.

2.2 How the pattern was found

We found the pattern in stages. Our first candidate reused the $D = 2$ formula one Hasse step lower. It failed in all 1,593 ordinary, nondegenerate $D = 0$ cells. This negative control showed that a new first-order correction was needed.

Reflection to the complementary wedge then suggested replacing the earlier quadratic $i'^2 + (k - 1)i' - n(n + 1)$ by $q_1(n, i')$. On the 692 cross-wedge cells, the resulting class agreed with

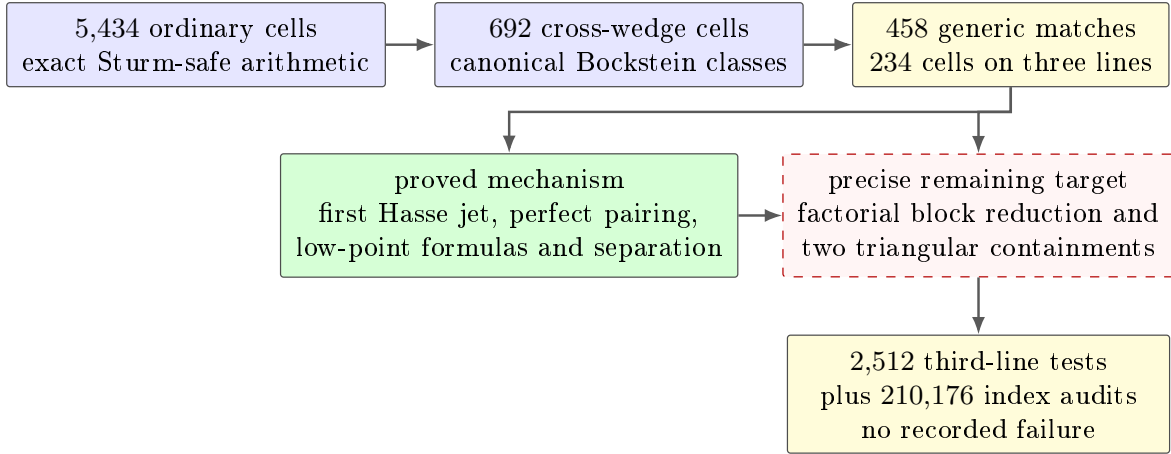


Figure 1: The experiment-to-theorem path. Exact computation first isolated the generic quadratic and its three boundary lines. The green box records the uniform results proved here. The dashed box is the remaining symbolic identity; finite tests support it but do not prove it.

the direct Bockstein in 458 cases. The 234 disagreements did not scatter: they were exactly the cells on the three affine lines

$$i' = n + 2, \quad i' + n = 2p - k - 1, \quad i' + n = 2p - k.$$

Each line contains 78 cells. Away from these lines, the direct class vanished in exactly 20 cases, and these were exactly the cases $q_1(n, i') = 0$. Thus the shifted conic was found only after the first formula failed and the full residuals identified the three boundary cases.

The boundary residuals suggested further exact formulas. On $i' = n + 2$, the residual always lay in one explicit block direction. Rational reconstruction across primes gave the scalar

$$-m(k - 2m - 2)!, \quad m = p - 2 - n.$$

On the two triangular lines, exact elimination placed every residual in an explicit triangular span. It also gave the endpoint coefficient $-(m + 1)$. These findings led to the Hasse-jet and filtration arguments proved below.

2.3 Validation and negative controls

Table 1 separates discovery calculations, blind extensions, independent form tests, and symbolic-index audits. A zero failure count means that the exact identity held over the stated finite range. It is not a theorem for arbitrary primes.

The relevant cusp-form space in the weight-24 experiment has dimension two. The experiment tests two generators and their projective linear combinations. It therefore does more than repeat the calculation on one one-dimensional family. The third-line extension is also separate from the original 78 boundary cells. It tests new primes and weights after the one-block shape and factorial scalar have been fixed.

2.4 What would falsify the conjectural mechanism

Each computational claim has a direct failure test. Any of the following would disprove part of the proposed mechanism:

1. an interior cross-wedge cell with $\mathcal{B}_i \neq q_1(n, i')X^p\theta^{i-p}f$;

Audit	Exact range	Outcome
Ordinary-cell census	5,434 cells	Establishes the ambient finite record and its admissible subfamilies.
Naive lower-block control	1,593 $D = 0$ cells	The reused $D = 2$ formula failed in all cells, isolating the missing projection term.
Cross-wedge discovery	692 cells	458 interior matches, 234 boundary disagreements, and 20 shifted-conic zeros.
Explicit first-Hasse-jet lift	The same 692 cells	No weight, mod- p , or canonical-Bockstein mismatch.
Third-line blind extension	652 cells through $p = 79$	No zero residual, block-shape failure, or factorial-scalar mismatch.
Independent weight-24 forms	186 ordinary forms and 1,860 tests through $p = 43$	No zero residual, block-shape failure, or scalar mismatch.
Triangular endpoints	156 boundary cells	Endpoint rank increased by one and the coefficient was $-(m + 1)$ in every case.
Index and filtration audit	210,176 parameter cells through $p = 101$	No exponent, weight, pairing, or filtration-gap failure.
Affine-reduction audit	614 factorial-range cells and 156 recurrence steps	No coefficient, sign, or low-point branch failure.

Table 1: Exact audits used in the discovery and validation of the proposed shifted-conic mechanism. The two large validation experiments were written as falsification tests: they count zero residuals, shape failures, and scalar mismatches separately.

2. a cell with $q_1(n, i') = 0$ and nonzero interior Bockstein, or an interior zero with $q_1(n, i') \neq 0$;
3. a third-line residual outside the predicted one-block direction;
4. a third-line scalar different from $-m(k - 2m - 2)!$ or dependent on the chosen ordinary form;
5. a triangular residual outside the stated span or with endpoint coefficient different from $-(m + 1)$.

The released scripts report each failure type separately. The tests remain useful even though the factorial block reduction is still open.

3 Notation and proof roadmap

3.1 Core notation

This section collects the symbols used throughout the proof. We include the band level $D(i)$ so that later references to the $D = 0$ and $D = 2$ strata are self-contained.

Symbol	Meaning
p, k, f	$p \geq 5$ is prime; $f \in M_k(\mathbb{Z}_{(p)})$ is nonzero modulo p , has $\omega_p(f) = k$, and is ordinary at p , with $4 \leq k < p - 1$.
θ, ω_{p^a}	The operator $q d/dq$ and weight filtration modulo p^a , respectively.
$\delta(\theta^i f)$	The integer defined on the admissible cells by $\omega_{p^2}(\theta^i f) = k + 2i + \delta(\theta^i f)P$.
$i = np + i'$	Band index, with n the segment number and i' the position inside the segment.
P	The full Hasse drop $p(p - 1)$.
$D(i)$	The least integer D such that $k + 2i + DP \geq \omega_p(\theta^{i-p} f) + p(p + 1)$; this is the inherited band upper-bound level.
w_-	The candidate lower weight $k + 2i - P$ in the $D = 0$ stratum.
$A, \tilde{A}, S, Z$	A is the Hasse invariant over $\mathbb{F}_p$, $\tilde{A} = E_{p-1}$ is its chosen integral lift, $S = \text{div}(A)$, and $Z = pS$ is its length- p thickening. In a purely integral formula, A may be used for E_{p-1} after this is stated explicitly.
Q_r	The Hasse quotient $H^0(Z, \omega^r _Z)$. Restriction gives an injection $M_r(\mathbb{F}_p)/A^p M_{r-P}(\mathbb{F}_p) \hookrightarrow Q_r$; it is an isomorphism when the restriction map is onto.
$\mathcal{B}_i, \mathcal{L}_i$	The Bockstein class of $\theta^i f$ at weight w_- and its shifted leading-block candidate. Their difference is the boundary residual.
X	The divided Eisenstein series $E_2/12$ modulo p ; powers of X form the direct block basis used in the reduction.
α, Φ_r, η_r	The first Hasse jet $(A - 1)/p$, the periodicity quotient $(\theta^{r+p-1} f - \theta^r f)/p$, and the lift error $(G_r - \theta^r f)/p$, all reduced modulo p .
Z_d	The block direction $X^d \theta^{p-k} f$ in the factorial reduction.
$i^\dagger$	The complementary index $p^2 + 1 - k - i$ used in the Tate-dual comparison.

Terminology. Write $f = \sum a(n)q^n$. Following Ahlgren–Raum–Richter [2], f is *non-ordinary* at p if $a(np) \equiv 0 \pmod{p}$ for every n , or equivalently if $\theta^{p-1} f \equiv f \pmod{p}$. It is *ordinary* otherwise. This definition applies to all modular forms, not only to eigenforms.

A *first Hasse jet* is a class on the thickened Hasse divisor Z . A *Bockstein class* is the intrinsic obstruction to an integral weight-lowering lift. A *minimal-weight projection* is a chosen characteristic-zero representative of the smallest candidate weight. The *block basis* is the direct basis built from powers of X and theta iterates. A *boundary residual* is the difference between the Bockstein class and its shifted leading-block candidate on one of the three affine boundary lines.

An *admissible* $D = 0$ – $D = 2$ *cross-wedge cell* means that the lower index has $D(i) = 0$, its Tate complement has $D(i^\dagger) = 2$, and there are integers m, d such that

$$n = p - 2 - m, \quad i' = p - k + m + 1 + d,$$

with

$$1 \leq m \leq \left\lfloor \frac{k-4}{2} \right\rfloor, \quad 0 \leq d \leq k - 2m - 1.$$

These are the cross-wedge inequalities used below. In particular, every factorial and every power in the uniform statements has a nonnegative exponent.

3.2 Dependency roadmap

The proof has five separate layers. The first three hold for all admissible parameters and do not use the conjectural scalar identity.

Layer	Main result	Logical role
Geometric	Perfect Hasse-quotient duality and theta skew-adjointness	Uses duality on Z ; independent of all finite computations and block formulas.
Integral	Bockstein realization, Hasse inclusion, and the explicit periodic first-Hasse-jet formula	Uses integral Eisenstein lifts and the first ordinary low-point congruence; supplies the minimal-weight projection and closed formulas for both low-point jet corrections.
Filtration	Uniform separation of all three boundary directions and nonvanishing of the factorial scalar candidate	Uses exact mod- p filtration; independent of the factorial block reduction.
Computational	Exact classification of 692 cross-wedge cells and 2,512 third-line stress tests	Proved only on the deposited finite range; discovers and falsifies formulas but is not extrapolated to all p .
Conditional	Factorial block reduction and two triangular containments for the endpoint core	Would identify the actual boundary residuals and imply the shifted-conic filtration law.

Thus the dependency chain is

$$\begin{array}{c}
\boxed{\text{geometry}} + \boxed{\text{integral first Hasse jet}} \implies \boxed{\text{explicit Bockstein class}}, \\
\boxed{\text{explicit Bockstein class}} \xrightarrow[\text{open}]{\text{block reduction}} \boxed{\text{shifted-conic law}}.
\end{array}$$

The boundary-separation theorem branches from the filtration layer before the open implication and therefore remains unconditional.

4 Bocksteins and Hasse quotients

4.1 The weight-lowering Bockstein

Let $R = \mathbb{Z}_{(p)}$, and let $M_w(R)$ denote the space of level-one modular forms of weight w with coefficients in R . We identify a form with its q -expansion. Write

$$\overline{M}_w := M_w(R)/pM_w(R)$$

and let $\overline{\mathcal{D}}$ be the space of divided congruences modulo p , viewed through their q -expansions.

Definition 4.1. Let $g \in \mathcal{D}(R/p^2R)$ and suppose that $g \pmod{p}$ is represented by some $h \in M_w(R)$. Define

$$\beta_w(g) := \left[\frac{g-h}{p} \right] \in \overline{\mathcal{D}}/\overline{M}_w.$$

Here any lift of the q -expansion of g to R may be used in the numerator.

We call w an *admissible comparison weight* for g if $g \pmod{p}$ has a representative in $M_w(R)$ and every smaller candidate weight for a mod- p^2 representative is a weight w' for which $w - w'$ is a nonnegative multiple of $P = p(p-1)$. This is the weight class used in the cross-wedge calculation.

Lemma 4.2. The class $\beta_w(g)$ is independent of the choices in Definition 4.1. Moreover,

$$\beta_w(g) = 0 \iff g \equiv H \pmod{p^2} \text{ for some } H \in M_w(R).$$

Consequently, if w is an admissible comparison weight for g , then

$$\beta_w(g) = 0 \iff \omega_{p^2}(g) \leq w.$$

Proof. Suppose first that $h, h' \in M_w(R)$ both represent $g \pmod{p}$. Then $h - h'$ has q -expansion divisible by p . The q -expansion lattice $M_w(R)$ is saturated: an integral modular form whose coefficients are all divisible by p belongs to $pM_w(R)$. One can also see this directly by forward substitution in the standard integral echelon basis, whose pivot coefficients are units. Thus $h - h' \in pM_w(R)$. Hence

$$\frac{g - h}{p} - \frac{g - h'}{p} = \frac{h' - h}{p} \in M_w(R),$$

so the two choices define the same class modulo $\overline{M}_w$. Changing the integral lift of g changes the quotient by an arbitrary multiple of $p^2/p = p$, and hence does not change its reduction modulo p . This proves well-definedness.

If $\beta_w(g) = 0$, there is an $m \in M_w(R)$ such that $(g - h)/p \equiv m \pmod{p}$. Thus

$$g \equiv h + pm \pmod{p^2},$$

and $h + pm$ belongs to $M_w(R)$. Conversely, suppose that $g \equiv H \pmod{p^2}$ for some $H \in M_w(R)$. Since $H \equiv h \pmod{p}$, the same echelon-basis argument gives $H - h = pm$ for some $m \in M_w(R)$. It follows that

$$\frac{g - h}{p} \equiv \frac{H - h}{p} = m \pmod{p},$$

so $\beta_w(g) = 0$ in $\overline{\mathcal{D}}/\overline{M}_w$. For the final equivalence, assume that every smaller admissible representative weight differs from w by a nonnegative multiple of $P = p(p - 1)$. Multiplication by the corresponding power of $E_{p-1}^p \equiv 1 \pmod{p^2}$ raises such a representative to weight w without changing its reduction modulo p^2 . Therefore a representative of weight at most w exists exactly when one of weight w exists. This is the asserted filtration criterion. $\square$

Remark 4.3. The canonical echelon remainder used in the computations is a coordinate representative of $\beta_w(g)$. Reducing g modulo the integral echelon basis of $M_w(R)$ kills the same leading coefficients as the lift h in the proof. The resulting remainder is divisible by p , and division by p followed by reduction modulo p gives the displayed class. Thus the computational zero test is intrinsic even though its coordinate vector depends on the chosen echelon basis.

4.2 Hasse duality and the Eisenstein bridge

Let $p \geq 5$ and put $P = p(p - 1)$. We first work on a compactified Katz modular curve X with auxiliary full level 3. Let ω be the Hodge bundle, and let C be the cusp divisor. Let $A \in H^0(X, \omega^{p-1})$ be the Hasse invariant. Its zero divisor S is reduced and disjoint from C . Put $Z = pS$. Thus Z is the Cartier divisor cut out by A^p . For every integer r , define

$$\mathcal{Q}_r := \omega^r / A^p \omega^{r-P} \simeq \omega^r|_Z, \quad Q_r := H^0(Z, \mathcal{Q}_r).$$

All schemes, line bundles, and sections in this subsection are over $\mathbb{F}_p$ unless we state another base. Let π denote the tame full-level cover of the level-one compactified moduli stack. The following standard descent fact records exactly how we use this cover.

Lemma 4.4 (Faithful tame pullback). *Let $\mathcal{L}$ be a line bundle on the level-one stack, or on a closed subscheme obtained from it by base change. Pullback by π is injective on global sections. If the cover is Galois with group G , then its image is the G -invariant subspace. A G -invariant perfect pairing upstairs restricts to a perfect pairing on the invariant subspaces.*

Proof. The cover is finite, flat, and surjective, and its degree is prime to p . The trace satisfies $\text{Tr}_\pi \circ \pi^* = (\deg \pi) \text{id}$. Since $\deg \pi$ is a unit in $\mathbb{F}_p$, pullback is injective. The same argument applies after base change to a closed subscheme, including the thickened Hasse divisor. For a Galois cover, exactness of the Reynolds operator identifies downstairs sections with invariants. Finally, exactness of invariants gives $(V^\vee)^G \simeq (V^G)^\vee$ for every finite-dimensional G -module V . This proves the assertion about perfect pairings. $\square$

All constructions below commute with G . One can alternatively state them directly on the tame compactified moduli stack.

Proposition 4.5 (Perfect Hasse-quotient pairing). *For every r there is a canonical perfect pairing*

$$\langle \ , \ \rangle_r : Q_r \times Q_{P+2-r} \longrightarrow \mathbb{F}_p.$$

It is the product pairing followed by the trace map on the finite Gorenstein scheme Z .

Proof. Because Z is an effective Cartier divisor on the smooth curve X , its dualizing sheaf is

$$\omega_Z \simeq (\omega_X \otimes \mathcal{O}_X(Z))|_Z.$$

Kodaira–Spencer gives $\omega_X(C) \simeq \omega^2$. We also have $\mathcal{O}_X(Z) \simeq \omega^P$. The canonical section of $\mathcal{O}_X(C)$ is nonzero on Z because $C \cap Z = \emptyset$. Hence

$$\omega_Z \simeq \omega^{P+2}|_Z.$$

Multiplication now gives

$$Q_r \otimes Q_{P+2-r} \longrightarrow \omega_Z.$$

Apply the trace map $\mathrm{Tr}_Z : H^0(Z, \omega_Z) \rightarrow \mathbb{F}_p$ to the resulting global section. This gives the displayed pairing. Duality for the finite Gorenstein scheme Z identifies Q_{P+2-r} with $\mathcal{H}om_Z(Q_r, \omega_Z)$. Therefore the pairing is perfect. The level-one statement follows from Lemma 4.4. $\square$

Let Θ denote the Katz theta operator in characteristic p [9], normalized by its q -expansion

$$\Theta\left(\sum a_n q^n\right) = \sum n a_n q^n.$$

It raises weight by $p+1$, satisfies the product rule, and satisfies $\Theta(A) = 0$. Hence it induces

$$\Theta : Q_r \longrightarrow Q_{r+p+1}.$$

Lemma 4.6 (Trace and tame Igusa residues). *Let $s \in S$, and let t be a local equation for S at s . After a finite extension of the residue field, the completed Igusa cover has a parameter u with $t = u^{p-1}$. Suppose that a local section of ω_Z is represented by*

$$F(t) \frac{dt}{t^p},$$

Its local Grothendieck trace is the coefficient of t^{p-1} in $F(t)$, up to the fixed unit from the chosen trivialization. Its pullback has residue $(p-1)$ times this coefficient. Therefore an exact pullback has local trace zero.

Proof. For the Cartier divisor $t^p = 0$, the usual residue description of duality identifies the local trace with

$$\mathrm{res}_{t=0} F(t) \frac{dt}{t^p}.$$

Only the term t^{p-1} contributes. Pulling back by $t = u^{p-1}$ gives

$$(p-1)F(u^{p-1})u^{p-2-p(p-1)} du.$$

Its coefficient of $u^{-1}du$ is $(p-1)$ times the coefficient of t^{p-1} in $F(t)$. Since $p-1$ is nonzero in $\mathbb{F}_p$, a zero residue after pullback gives a zero local trace. The finite residue-field extension does not change this calculation. This follows from transitivity of the field trace. $\square$

Proposition 4.7 (Theta adjointness). *If $x \in Q_r$ and $y \in Q_{P+1-p-r}$, then*

$$\langle \Theta x, y \rangle_{r+p+1} = -\langle x, \Theta y \rangle_r.$$

More generally, for $m \geq 0$ and $y \in Q_{P+2-r-m(p+1)}$,

$$\langle \Theta^m x, y \rangle_{r+m(p+1)} = (-1)^m \langle x, \Theta^m y \rangle_r.$$

Proof. The two products in the first formula have weight $P+2$. Thus both traces are defined. The product rule gives

$$(\Theta x)y + x(\Theta y) = \Theta(xy).$$

Notice that $h := xy$ has weight

$$P+1-p = (p-1)^2.$$

Let a be the canonical section on the Igusa curve, normalized by $a^{p-1} = A$. Under Kodaira–Spencer, pull back the meromorphic differential represented by $\Theta h/A^p$. The result is the exact differential attached to the function $h/a^{(p-1)^2}$. This is the geometric description of theta in [7, Proposition 5.8].

We apply this formula to any local lift of h from Z to X . The answer does not depend on the lift. Indeed, another lift differs by $A^p g$. Its theta derivative differs by $A^p \Theta g$, which is zero on Z .

Now apply Lemma 4.6 at each supersingular point. Each local component of $\text{Tr}_Z(\Theta h)$ is zero. The sum of the local traces is therefore zero. Hence $\text{Tr}_Z(\Theta(xy)) = 0$. The product rule gives the first identity. Induction on m gives the second identity. At each step, the weight of y decreases by $p+1$. There is no cusp contribution: Z is supported on S , and $S \cap C = \emptyset$. $\square$

The weight relation relevant to the Tate complement contains a further shift of $p(p+1)$. Let

$$B := E_{p+1} \pmod{p} \in H^0(X, \omega^{p+1}).$$

Lemma 4.8 (The Eisenstein bridge is a unit). *The forms A and B have no common zero. Hence $B|_Z$ is a unit and multiplication by B^p gives an isomorphism*

$$B^p : Q_r \xrightarrow{\sim} Q_{r+p(p+1)}$$

for every r .

Proof. The Hasse invariant has a simple zero at every point of S . Let $\partial_k = \Theta - kE_2/12$ be the Serre derivative in weight k . The q -expansion of A is 1, so $\Theta(A) = 0$ in characteristic p . Also $E_2 \equiv E_{p+1} = B \pmod{p}$. Hence the q -expansion principle gives

$$12\partial_{p-1}A = B \quad \text{in characteristic } p.$$

At a zero of A , the connection term is a multiple of A and vanishes. Thus $\partial_{p-1}A$ is the ordinary derivative of a local equation for S , up to a nonzero change of trivialization. It is nonzero because the zero is simple. Therefore B does not vanish on S . Its restriction to the Artinian thickening Z is consequently invertible, and so is B^p . $\square$

Theorem 4.9 (Perfect Tate-complement pairing). *For every r the formula*

$$[x, z]_r := \langle x, B^{-p}z \rangle_r$$

defines a canonical perfect pairing

$$[\ , \]_r : Q_r \times Q_{P+2+p(p+1)-r} \longrightarrow \mathbb{F}_p.$$

Under this pairing theta is skew-adjoint. More precisely, if

$$z \in Q_{P+1-p+p(p+1)-r},$$

then

$$[\Theta x, z]_{r+p+1} = -[x, \Theta z]_r.$$

The same identity iterates with the sign $(-1)^m$ for Θ^m .

Proof. Lemma 4.8 identifies the second factor with Q_{P+2-r} . Perfectness now follows from Proposition 4.5. Theta is a derivation in characteristic p , so

$$\Theta(B^p) = pB^{p-1}\Theta(B) = 0, \quad \Theta(B^{-p}) = 0$$

on Z . Proposition 4.7 therefore gives

$$\begin{aligned} [\Theta x, z]_{r+p+1} &= \langle \Theta x, B^{-p}z \rangle_{r+p+1} \\ &= -\langle x, \Theta(B^{-p}z) \rangle_r \\ &= -\langle x, B^{-p}\Theta z \rangle_r = -[x, \Theta z]_r. \end{aligned}$$

Iteration proves the final assertion. □

Now let

$$w_- = k + 2i - P, \quad w_+ = k + 2i^\dagger + P, \quad i + i^\dagger = p^2 + 1 - k.$$

Then

$$w_- + w_+ = 2p^2 + 2 = P + 2 + p(p + 1),$$

and therefore

$$[\ , \]_{w_-} : Q_{w_-} \times Q_{w_+} \longrightarrow \mathbb{F}_p$$

is a perfect pairing under which theta is skew-adjoint. This is the pairing at the two candidate-weight indices. The factor B^p has weight $p(p + 1)$. It is the same factor that occurs in the leading-block formula for the $D = 2$ obstruction. Serre duality alone pairs weights whose sum is $P + 2$. The Eisenstein bridge gives the extra shift that we need.

There is one important indexing point. The obstruction to lowering to a candidate weight w lies in Q_{w+P} , not in Q_w . This is proved in Proposition 4.10. The pairing above is therefore a structural pairing. It is not yet the pairing of the two actual Bockstein target spaces. Theorem 4.13 constructs that pairing with the correct shift. This distinction matters: a constant sum of candidate weights does not by itself prove Bockstein compatibility.

4.3 Geometric realization of the weight-lowering obstruction

Retain the notation of the Hasse-duality subsection. In particular, $P = p(p - 1)$, $S = \text{div}(A)$, $Z = pS$, and $Q_r = H^0(Z, \omega^r|_Z)$. Let $\tilde{A} = E_{p-1}$ be the normalized integral lift of the Hasse invariant. Its q -expansion is congruent to 1 modulo p , and therefore

$$\tilde{A}^p \equiv 1 \pmod{p^2} \tag{2}$$

as a q -expansion.

Proposition 4.10 (The Bockstein class is a first Hasse jet). *Let g be a divided congruence modulo p^2 . Suppose that*

1. *g has a representative $G \in M_{w+P}(\mathbb{Z}_{(p)})$ modulo p^2 ; and*
2. *$g \bmod p$ has a representative $h \in M_w(\mathbb{Z}_{(p)})$.*

Then

$$\Gamma_w(g) := \left(\frac{G - \tilde{A}^p h}{p} \bmod p \right) \Big|_Z \in Q_{w+P}$$

is independent of all choices. Moreover,

$$\Gamma_w(g) = 0 \iff g \text{ has a representative in } M_w(\mathbb{Z}_{(p)}) \text{ modulo } p^2.$$

Under the q -expansion identification, $\Gamma_w(g)$ is the geometric image of the divided-congruence class $\beta_w(g)$ from Lemma 4.2.

Proof. The numerator is divisible by p : both terms have weight $w + P$, and their q -expansions agree modulo p because $\tilde{A}^p \equiv 1 \pmod{p}$. Thus the displayed quotient reduces to a form of weight $w + P$ modulo p .

Changing G without changing its q -expansion modulo p^2 changes the quotient by zero modulo p , by the saturated q -expansion principle. If h' is another choice, the integral echelon argument used in Lemma 4.2 gives $h' - h = pm$ for some $m \in M_w$. The two quotients then differ modulo p by

$$-A^p m \in A^p M_w.$$

Its restriction to Z is zero. This proves independence.

The restriction sequence for the Cartier divisor Z is

$$0 \longrightarrow \omega^w \xrightarrow{A^p} \omega^{w+P} \longrightarrow \omega^{w+P}|_Z \longrightarrow 0.$$

Exactness at $H^0(X, \omega^{w+P})$ says that the kernel of restriction is exactly $A^p M_w$; no surjectivity assertion is needed here. Hence $\Gamma_w(g) = 0$ exactly when

$$\frac{G - \tilde{A}^p h}{p} \equiv A^p m \pmod{p}$$

for some $m \in M_w$. In that case

$$G \equiv \tilde{A}^p(h + pm) \pmod{p^2}.$$

By (2), the q -expansion of the right side is that of $h + pm$ modulo p^2 , so $h + pm$ is a weight- w representative of g . The converse follows by reversing the argument. Finally, before restriction to Z the displayed quotient has the same q -expansion as $(g - h)/p$ modulo p . The restriction map induces an injection

$$M_{w+P}(\mathbb{F}_p)/A^p M_w(\mathbb{F}_p) \hookrightarrow Q_{w+P},$$

and $\Gamma_w(g)$ is the image of the divided-congruence class under this map. This proves the last assertion. $\square$

The index shift in Proposition 4.10 is essential: the obstruction to lowering to weight w belongs to Q_{w+P} , not to Q_w .

Lemma 4.11 (Theta commutes with the integral Hasse inclusion). *For every integral q -series h and every $m \geq 0$,*

$$\theta^m(\tilde{A}^p h) \equiv \tilde{A}^p \theta^m(h) \pmod{p^2}.$$

Proof. Write $\tilde{A} = 1 + pa(q)$. Then $\theta\tilde{A}$ is divisible by p , and hence

$$\theta(\tilde{A}^p) = p\tilde{A}^{p-1}\theta\tilde{A} \equiv 0 \pmod{p^2}.$$

The product rule gives the assertion for $m = 1$, and iteration proves it for every m . $\square$

Lemma 4.12 (A horizontal unit of the required weight). *After pullback to a compactified modular curve with auxiliary full level 3, there is a form*

$$F \in H^0(X, \omega^{3p-1})$$

whose restriction to S is nowhere zero. Consequently

$$U := F^p|_Z \in H^0(Z, \omega^{p(3p-1)}|_Z)$$

is a unit and $\Theta(U) = 0$.

Proof. Multiplication by A gives

$$0 \longrightarrow \omega^{2p} \xrightarrow{A} \omega^{3p-1} \longrightarrow \omega^{3p-1}|_S \longrightarrow 0.$$

Serre duality and Kodaira–Spencer give

$$H^1(X, \omega^{2p})^\vee \simeq H^0(X, \omega^{2-2p}(-C)) = 0.$$

The restriction map onto S is therefore surjective. Since S is a finite reduced scheme, its coordinate ring is a product of finite fields, and every line bundle on it has a global generator. Lift such a generator to F . At every point of Z , its reduction modulo the nilpotent ideal is a generator; Nakayama’s lemma therefore shows that F generates on Z . Its p -th power is a unit of weight $p(3p-1)$, and the product rule in characteristic p gives $\Theta(F^p) = 0$. $\square$

Now put

$$w_- = k + 2i - P, \quad w_+ = k + 2i^\dagger + P,$$

and define the actual geometric Bockstein weights

$$r_- := w_- + P = k + 2i, \quad r_+ := w_+ + P = k + 2i^\dagger + 2P.$$

They satisfy

$$r_- + r_+ = P + 2 + p(3p-1). \tag{3}$$

Theorem 4.13 (Auxiliary-level natural Bockstein pairing). *Choose F as in Lemma 4.12 and put $W = p(3p-1)$. For every integer r , the formula*

$$[x, z]_{r,F}^{\text{nat}} := \langle x, U^{-1}z \rangle_r$$

defines a perfect pairing

$$Q_r \times Q_{P+2+W-r} \longrightarrow \mathbb{F}_p.$$

If $x \in Q_r$ and $z \in Q_{P+2+W-r-(p+1)}$, then the correctly typed adjointness identity is

$$[\Theta x, z]_{r+p+1,F}^{\text{nat}} = -[x, \Theta z]_{r,F}^{\text{nat}}.$$

Taking $r = r_-$ gives a perfect pairing $Q_{r_-} \times Q_{r_+} \rightarrow \mathbb{F}_p$. The construction is not canonical in F , but every choice is perfect and detects the zero class. Pullback to auxiliary level is faithful, so a level-one Bockstein vanishes if and only if its pulled-back class vanishes.

Proof. The unit U has weight W . Its inverse identifies $Q_{P+2+W-r}$ with Q_{P+2-r} . Perfectness therefore follows from Proposition 4.5. Since $\Theta(U) = \Theta(U^{-1}) = 0$, Proposition 4.7 gives

$$\begin{aligned} [\Theta x, z]_{r+p+1,F}^{\text{nat}} &= \langle \Theta x, U^{-1}z \rangle_{r+p+1} \\ &= -\langle x, \Theta(U^{-1}z) \rangle_r \\ &= -\langle x, U^{-1}\Theta z \rangle_r = -[x, \Theta z]_{r,F}^{\text{nat}}. \end{aligned}$$

Equation (3) gives $r_+ = P + 2 + W - r_-$, so the specialization at r_- has the stated factors. Lemma 4.4, applied after base change to the thickened Hasse divisor, proves the final assertion. $\square$

Remark 4.14. The family of perfect pairings in Theorem 4.13 depends on the choice of the auxiliary-level unit F . The theorem uses it only to identify dual vector spaces and detect the zero class after faithful pullback. It does not assert a canonical level-one bilinear form or descend the chosen unit through the tame deck group. By contrast, the unshifted Hasse-quotient pairing in Proposition 4.5 is canonical and does descend by invariants.

For the complementary theta iterates, Proposition 4.10 now gives the correctly typed classes

$$\Gamma_{w_-}(\theta^i f) \in Q_{r_-}, \quad \Gamma_{w_+}(\theta^{i^\dagger} f) \in Q_{r_+}.$$

The remaining cross-wedge problem is no longer the construction of their ambient pairing. It is the following concrete compatibility statement: under the isomorphism $Q_{r_+} \simeq Q_{r_-}^\vee$ induced by $[\ , \]^{\text{nat}}$, the second displayed class must equal the functional obtained from the first by the integral theta-iterate construction, up to the explicit nonzero scalar in the $D = 2$ block formula. Perfectness and theta adjointness alone do not prove that identification.

Lemma 4.11 proves the Hasse-inclusion part of the required square. Theorem 5.2 gives an explicit formula for the lower minimal-weight projection. The unresolved part is its factorial block reduction and the corresponding compatibility with the complementary class.

4.4 An abstract Bockstein-adjointness criterion

The final compatibility can be reduced to a standard algebraic check. We record the check separately so that the remaining modular input is visible.

Lemma 4.15 (Adjoint Bocksteins). *Let $R = \mathbb{Z}/p^2\mathbb{Z}$. Let $C^\bullet$ and $D^\bullet$ be bounded complexes of finite free R -modules with a perfect pairing*

$$C^a \times D^{N-a} \longrightarrow R$$

for which the differentials are graded skew-adjoint. Let

$$\begin{aligned} \partial_C : H^a(C^\bullet/p) &\longrightarrow H^{a+1}(C^\bullet/p), \\ \partial_D : H^{N-a-1}(D^\bullet/p) &\longrightarrow H^{N-a}(D^\bullet/p). \end{aligned}$$

be the connecting maps from multiplication by p . Then

$$\langle \partial_C x, y \rangle = (-1)^{a+1} \langle x, \partial_D y \rangle.$$

Proof. Choose cocycle lifts $\tilde{x}$ and $\tilde{y}$ over R . Write $d\tilde{x} = px_1$ and $d\tilde{y} = py_1$. Graded skew-adjointness gives

$$\langle d\tilde{x}, \tilde{y} \rangle = (-1)^{a+1} \langle \tilde{x}, d\tilde{y} \rangle.$$

Divide by p and reduce modulo p . The resulting classes of x_1 and y_1 are the two connecting maps, which proves the formula. $\square$

To apply Lemma 4.15 here, one must construct a self-dual two-term complex whose connecting maps are exactly the geometric classes $\Gamma_{w_-}(\theta^i f)$ and $\Gamma_{w_+}(\theta^{i^\dagger} f)$. Proposition 4.10 supplies the target spaces, and Theorem 4.13 supplies the duality. The unresolved input is commutativity of the integral theta-iterate square with the two minimal-weight projections. Lemma 4.11 proves commutativity with the E_{p-1}^p weight-raising maps. What remains is the projection term introduced by integral theta modulo p^2 , including its unit scalar. Verifying that term would turn the abstract lemma into the desired cross-wedge theorem.

5 Direct calculation in the lower subwedge

The geometric pairing isolates the correct obstruction spaces, but the auxiliary horizontal unit in the natural pairing is noncanonical. For the vanishing theorem there is a more concrete route: calculate the lower Bockstein directly from the E_2 expansion, in the style of [2].

Write $i = np + i'$ and put

$$q_1(n, i') = i'^2 + (k-1)i' - (n+1)(n+2) \in \mathbb{F}_p, \quad X = E_2/12 \pmod{p}.$$

For a $D = 0$ cell let

$$w_- = k + 2i - P, \quad \mathcal{B}_i = \Gamma_{w_-}(\theta^i f) \in Q_{k+2i}.$$

On the $D = 0$ stratum used here, the integral theta expansion of [2, 3] gives a representative of $\theta^i f$ in $M_{k+2i}(\mathbb{Z}_{(p)})$ modulo p^2 . Thus Proposition 4.10 applies and the displayed class is well defined. The shifted leading-block candidate is the image in the same quotient of

$$\mathcal{L}_i = q_1(n, i') X^p \theta^{i-p} f. \quad (4)$$

Proposition 5.1 (Finite exact audit). *In the current Sturm-safe record of 692 ordinary, non-degenerate $D = 0$ - $D = 2$ complementary pairs, the following statements hold exactly.*

1. *Away from the three affine lines*

$$i' = n + 2, \quad i' + n = 2p - k - 1, \quad i' + n = 2p - k, \quad (5)$$

one has $\mathcal{B}_i = \mathcal{L}_i$ in all 458 cells.

2. *On the three lines in (5), the equality fails in all 234 cells, but $\mathcal{B}_i$ is nonzero in every cell.*
3. *The 20 vanishing classes occur off the three lines and are exactly the cells with $q_1(n, i') = 0$.*
4. *On the line $i' = n + 2$, all 78 tested residuals satisfy*

$$\mathcal{B}_i - \mathcal{L}_i = c_{p,k,n} X^{2n-2p+k+3} \theta^{p-k} f \quad \text{in } Q_{k+2i}, \quad (6)$$

for an explicitly computed scalar $c_{p,k,n} \in \mathbb{F}_p^\times$.

5. *Put $m = p - 2 - n$. On the line $i' + n = 2p - k - 1$, all 78 tested residuals lie in the explicit triangular span*

$$\left\langle X^a \theta^{2p-k-1-a} f : p-k \leq a \leq p-k+2m+1; X^p \theta^{p-k-1} f \right\rangle. \quad (7)$$

On the adjacent line $i' + n = 2p - k$, all 78 tested residuals lie in

$$\left\langle X^a \theta^{2p-k-a} f : p-k+1 \leq a \leq p-k+2m+3; X^p \theta^{p-k} f \right\rangle. \quad (8)$$

Let V_m^- and V_m^+ denote the spans in (7) and (8), respectively, after omitting the displayed X^p endpoint. In every tested cell the endpoint class is independent of the corresponding $V_m^\pm$, and

$$\mathcal{B}_i - \mathcal{L}_i \equiv -(m+1) X^p \theta^{p-k-1} f \pmod{V_m^-}, \quad i' + n = 2p - k - 1, \quad (9)$$

$$\mathcal{B}_i - \mathcal{L}_i \equiv -(m+1) X^p \theta^{p-k} f \pmod{V_m^+}, \quad i' + n = 2p - k. \quad (10)$$

Consequently every one of these 156 residuals is nonzero.

These statements are finite computations, not assertions for arbitrary p , k , and f .

Proof. The direct class is obtained by reducing $\theta^i f$ modulo p^2 by the echelon basis of M_{w-} , dividing the canonical remainder by p , and then reducing modulo p . The candidate in (4) is reduced by the same basis modulo p . Full Sturm-safe comparison gives 458 equal and 234 unequal classes. The unequal cells are classified by (5) with confusion counts

$$\text{TP} = 234, \quad \text{FP} = 0, \quad \text{FN} = 0, \quad \text{TN} = 458.$$

The direct remainder vanishes in precisely 20 cells; each has $q_1 = 0$. The boundary residuals are nonzero. A second full-remainder comparison tests (6) in all 78 cells on $i' = n + 2$; every residual is proportional to the displayed block and every scalar is nonzero. This verifies the proposition over the stated record. Finally, exact Gaussian elimination over $\mathbb{F}_p$ on the complete remainders places every residual on the other two lines in the spans (7) and (8). The restricted echelon solution has support equal to its column rank in every case. Omitting the final X^p column lowers the rank by one in all 156 cases, while the coefficient of that final column is $-(m + 1)$ in every case. Since $1 \leq m < p - 1$, this is a nonzero scalar and proves the finite nonvanishing assertions (9) and (10). $\square$

Theorem 5.2 (Explicit periodic first-Hasse-jet formula). *Assume that $f \in M_k(\mathbb{Z}_{(p)})$ is nonzero modulo p , is ordinary, has $\omega_p(f) = k$, and that $4 \leq k < p - 1$, and that (n, i') is an admissible $D = 0 - D = 2$ cross-wedge cell. Write*

$$n = p - 2 - m, \quad i' = p - k + m + 1 + d, \quad 0 \leq d \leq k - 2m - 1.$$

Put

$$r = p - k + d, \quad L = n + 1, \quad \ell_0 = p - k + 1.$$

Then $i = np + i' = r + L(p - 1)$.

Let $A = E_{p-1}$ and $B = E_{p+1}$ be their standard p -integral Eisenstein lifts, and, for $g \in M_v(\mathbb{Z}_{(p)})$, define

$$\mathcal{T}_v(g) = A\theta g + \frac{v}{12}(B - AE_2)g \in M_{v+p+1}(\mathbb{Z}_{(p)}). \quad (11)$$

Let $F_j(f)$ be the modified Serre derivatives, defined by

$$F_0(f) = f, \quad F_1(f) = \partial_k f, \quad F_{j+1}(f) = \partial_{k+2j} F_j(f) - \frac{j(j+k-1)}{144} E_4 F_{j-1}(f),$$

where $\partial_v g = \theta g - (v/12)E_2 g$. At the first ordinary low point, the Chida–Kaneko congruence [4, Lemma 2.4 and Proposition 2.5] gives

$$F_{\ell_0}(f) \in M_{k+2\ell_0}(\mathbb{Z}_{(p)}), \quad F_{\ell_0}(f) \equiv \theta^{\ell_0} f \pmod{p}.$$

Define a modular lift G_r of $\theta^r f$ as follows. If $r < \ell_0$, start with $G_0 = f$ and apply (11) r times. If $r \geq \ell_0$, start with $G_{\ell_0} = F_{\ell_0}(f)$ and apply (11) $r - \ell_0$ times. Its weight is

$$v_r = \begin{cases} k + r(p + 1), & r < \ell_0, \\ k + r(p + 1) - \ell_0(p - 1), & r \geq \ell_0. \end{cases}$$

If

$$a = \frac{w_- - v_r}{p - 1} = \begin{cases} k - 2m - 2, & d = 0, \\ p - 2m - d - 1, & d \geq 1, \end{cases}$$

then $a \geq 0$ and

$$H_i = A^a G_r \in M_{w-}(\mathbb{Z}_{(p)}), \quad H_i \equiv \theta^i f \pmod{p}. \quad (12)$$

In $\mathbb{F}_p[[q]]$, put

$$\begin{aligned}\alpha &= \frac{A-1}{p} \pmod{p}, \\ \Phi_r(f) &= \frac{\theta^{r+p-1}f - \theta^r f}{p} \pmod{p}, \\ \eta_r(f) &= \frac{G_r - \theta^r f}{p} \pmod{p}.\end{aligned}$$

These quotients are integral. The lower Bockstein is the class

$$\boxed{\mathcal{B}_i = [L\Phi_r(f) - \eta_r(f) - a\alpha\theta^r f]_{Q_{k+2i}}.} \quad (13)$$

Thus the minimal-weight projection correction is an explicit first-Hasse-jet operator; it is not an unspecified choice of characteristic-zero lift.

Proof. The Serre derivative identity rewrites (11) as

$$\mathcal{T}_v(g) = A\left(\theta g - \frac{v}{12}E_2g\right) + \frac{v}{12}Bg,$$

so it is modular of weight $v + p + 1$. The congruences $A \equiv 1$ and $B \equiv E_2 \pmod{p}$ give $\mathcal{T}_v(g) \equiv \theta g \pmod{p}$. Together with the Chida–Kaneko low-point congruence, induction gives $G_r \equiv \theta^r f \pmod{p}$ and the displayed formula for v_r . Substitution of $r = p - k + d$ and the two possible filtration branches gives the stated value of a . The cross-wedge inequalities make both formulas for a nonnegative. Hence (12) follows.

For a Fourier index j prime to p , write $j^{p-1} = 1 + pq_p(j) \pmod{p^2}$. Since $i = r + L(p-1)$,

$$j^i = j^r(1 + pLq_p(j)) \pmod{p^2}.$$

If $p \mid j$, then $r \geq p - k \geq 2$ and $i \geq r$. Hence both j^r and j^i are divisible by p^2 , so $(j^i - j^r)/p \equiv 0 \pmod{p}$. The same argument applies to $(j^{r+p-1} - j^r)/p$. Thus both sides of the corresponding divided coefficient identity vanish modulo p . Coefficientwise, therefore,

$$\frac{\theta^i f - \theta^r f}{p} \equiv L\Phi_r(f) \pmod{p}. \quad (14)$$

Also $A = 1 + p\alpha \pmod{p^2}$ and $G_r = \theta^r f + p\eta_r(f) \pmod{p^2}$, whence

$$H_i = A^a G_r \equiv \theta^r f + p(\eta_r(f) + a\alpha\theta^r f) \pmod{p^2}.$$

Subtracting this identity from (14), dividing by p , and passing to the lower Hasse quotient proves (13). $\square$

The first Hasse jet in (13) is itself recursive. Put

$$C = \frac{B - AE_2}{p} \pmod{p}.$$

Whenever $G_{t+1} = \mathcal{T}_{v_t}(G_t)$, direct expansion modulo p^2 gives

$$\eta_{t+1} = \theta\eta_t + \alpha\theta^{t+1}f + \frac{v_t}{12}C\theta^t f. \quad (15)$$

The initial jet is $\eta_0 = 0$ on the first branch and $(F_{\ell_0}(f) - \theta^{\ell_0}f)/p$ on the second. Equations (13) and (15) therefore reduce the remaining block identities to a finite symbolic calculation in three explicit divided q -series.

The jets at the two sides of the first low point can be controlled uniformly. This removes the jet comparison from the list of open claims.

Proposition 5.3 (The two low-point jets lie below the endpoint cutoffs). Put $s = p-k$, $h = s+1$, $U_t = \theta^t f$, and $\Xi = C/12$. Also put

$$S_s = \alpha U_s + X^p U_{s-1}, \quad Y = \Xi + X^p.$$

Then the following statements hold.

1. The pre-low-point jet satisfies

$$\eta_s + k S_s \in M_{k+s(p+1)}(\mathbb{F}_p). \quad (16)$$

On the first triangular line this space is contained in $M_{w_-}(\mathbb{F}_p)$.

2. The reset jet has the explicit formula

$$\eta_h = -h! \sum_{j=0}^{h-1} \frac{X^{h-j} U_j}{j!(h-j)} \quad \text{in } \mathbb{F}_p[[q]]. \quad (17)$$

It belongs to $M_{k+h(p+1)}(\mathbb{F}_p)$. On the second triangular line this space is contained in $M_{w_-}(\mathbb{F}_p)$.

Proof. For $r \geq 0$, write $(x)^{\bar{r}} = x(x+1) \cdots (x+r-1)$, with value 1 when $r = 0$. The recurrence defining the modified Serre derivatives, together with $\theta X = X^2 - E_4/144$, gives the triangular identities

$$U_t = \sum_{j=0}^t \binom{t}{j} (k+j)^{\overline{t-j}} X^{t-j} F_j, \quad (18)$$

$$F_t = \sum_{j=0}^t (-1)^{t-j} \binom{t}{j} (k+j)^{\overline{t-j}} X^{t-j} U_j. \quad (19)$$

The first formula follows by induction on t : the E_4 term produced by differentiating X^{t-j} cancels the E_4 term in the recurrence for F_{j+1} . The remaining X and F_{j+1} coefficients combine by Pascal's identity to give the coefficient at level $t+1$. The second formula is the inverse triangular relation. After substituting the first formula, use

$$\binom{t}{j} \binom{j}{a} = \binom{t}{a} \binom{t-a}{j-a}, \quad (k+a)^{\overline{j-a}} (k+j)^{\overline{t-j}} = (k+a)^{\overline{t-a}}.$$

The coefficient of U_a is then a fixed factor times $\sum_{j=a}^t (-1)^{t-j} \binom{t-a}{j-a}$. This is zero when $a < t$ and one when $a = t$.

Define

$$K_t = \sum_{j=0}^t \binom{t}{j} (k+j)^{\overline{t-j}} A^j (B/12)^{t-j} F_j.$$

Every summand has weight $k+t(p+1)$, and $K_t \equiv U_t \pmod{p}$. Since

$$A = 1 + p\alpha, \quad \frac{B}{12} = X + p(\alpha X + \Xi) \pmod{p^2},$$

coefficientwise expansion gives

$$\frac{K_t - U_t}{p} = t\alpha U_t + t(k+t-1)\Xi U_{t-1} \pmod{p}. \quad (20)$$

Here the coefficient of Ξ follows from

$$\sum_{j=0}^{t-1} \binom{t}{j} (k+j)^{\overline{t-j}} (t-j) X^{t-j-1} F_j = t(k+t-1) U_{t-1}.$$

Before the first low point, G_s and K_s are modular forms of the same weight $k+s(p+1)$, and both reduce to U_s . Hence $G_s - K_s$ has every Fourier coefficient divisible by p . The q -expansion lattice of level-one modular forms over $\mathbb{Z}_{(p)}$ is saturated, so $(G_s - K_s)/p$ reduces to a form in $M_{k+s(p+1)}(\mathbb{F}_p)$. Since $s = -k$ and $k+s-1 = -1$ in $\mathbb{F}_p$, equation (20) gives

$$\frac{K_s - U_s}{p} = -k\alpha U_s + k\Xi U_{s-1}.$$

Therefore

$$\eta_s + kS_s = \frac{G_s - K_s}{p} + kY U_{s-1} \pmod{p}.$$

Lemma 5.7 shows that Y has weight $p+1$. Also U_{s-1} is still before the first low point. Both terms on the right therefore have weight at most $k+s(p+1)$, proving (16). On the first triangular line,

$$w_- - (k+s(p+1)) = (k-2m-2)(p-1) > 0.$$

At the low point, $G_h = F_h$. In (19), every coefficient with $j < h$ contains the single factor p , because

$$(k+j)^{\overline{h-j}} = (k+j)(k+j+1) \cdots p.$$

Wilson's theorem gives

$$(-1)^{h-j} \binom{h}{j} \frac{(k+j)^{\overline{h-j}}}{p} = -\frac{h!}{j!(h-j)} \pmod{p}.$$

After subtracting U_h and dividing by p , this is (17). Every summand there has filtration at most $k+h(p+1)$. Finally, on the second triangular line,

$$w_- - (k+h(p+1)) = (k-2m-3)(p-1) > 0.$$

This proves both cutoff statements. □

The next reduction makes that calculation more precise. It is an exact identity of divided q -series. It does not use the conjectural block formula.

Proposition 5.4 (Affine reduction of the factorial target). *Keep the notation of Theorem 5.2, assume $d \geq 1$, and put*

$$s = p - k, \quad U_t = \theta^t f, \quad Z_d = X^d U_s.$$

Define

$$\begin{aligned} \mathcal{C}_d &= -\Phi_{s+d} - \eta_{s+d} + (d+1)\alpha U_{s+d} - d(d+1-k)X^p U_{s+d-1}, \\ \mathcal{M}_d &= -\Phi_{s+d} + 2\alpha U_{s+d} - (2d-k)X^p U_{s+d-1} + (d-1)!Z_d. \end{aligned}$$

Then

$$\mathcal{B}_i - \mathcal{L}_i + m(d-1)![Z_d] = [\mathcal{C}_d + m\mathcal{M}_d]_{Q_{k+2i}}. \quad (21)$$

In particular, the unresolved expression is affine in m before passage to the quotient. The quotient weight still depends on m , so this statement does not split the conjecture into two separate vanishing claims.

Proof. In $\mathbb{F}_p$ the parameters satisfy

$$L = p - 1 - m = -(m + 1), \quad a = p - 2m - d - 1 = -(2m + d + 1).$$

Direct substitution in the definition of q_1 gives

$$q_1(n, i') = m(2d - k) + d(d + 1 - k).$$

Also

$$i - p = (s + d - 1) + (L - 1)(p - 1).$$

Coefficientwise Fermat periodicity therefore gives $\theta^{i-p}f = U_{s+d-1}$ in $\mathbb{F}_p[[q]]$. Substituting these four identities into (13), subtracting $\mathcal{L}_i$, and adding $m(d - 1)![Z_d]$ gives exactly (21). $\square$

Lemma 5.5 (The first Hasse derivative). *In $\mathbb{F}_p[[q]]$ one has*

$$\theta\alpha = X - X^p. \quad (22)$$

Proof. The Eisenstein congruence of [1, Theorem 1.2], in the normalization used here, is

$$E_2 \equiv 12\partial A A^{2p-1} + pB^p A^{p-2} \pmod{p^2}, \quad \partial A = \theta A - \frac{p-1}{12}E_2 A.$$

Write $A = 1 + p\alpha$ modulo p^2 . Then

$$12\partial A = E_2 + p(12\theta\alpha + (\alpha - 1)E_2), \quad A^{2p-1} = 1 - p\alpha \pmod{p^2}.$$

After multiplication, the cited congruence reduces to

$$12\theta\alpha - E_2 + B^p = 0 \pmod{p}.$$

Finally, $B \equiv E_2 = 12X \pmod{p}$ and Fermat's theorem gives $B^p/12 = X^p$. Division by 12 proves (22). $\square$

The two affine pieces obey short recurrences. They isolate the exact identity that a uniform proof still has to control.

Proposition 5.6 (Recurrences for the affine pieces). *Put*

$$\Xi = \frac{C}{12} \in \mathbb{F}_p[[q]].$$

Whenever both sides are defined on the second branch of the ordinary theta cycle,

$$\mathcal{C}_{d+1} - \theta\mathcal{C}_d = ((k - d - 1)(\Xi + X^p) - (d + 1)X)U_{s+d}, \quad (23)$$

$$\mathcal{M}_{d+1} - \theta\mathcal{M}_d = -2XU_{s+d} + \frac{d!}{144}E_4X^{d-1}U_s - (d - 1)!X^dU_{s+1}. \quad (24)$$

These are identities in $\mathbb{F}_p[[q]]$, before taking a Hasse quotient.

Proof. The definition of Φ_t gives $\Phi_{t+1} = \theta\Phi_t$. On the second branch one has $v_t \equiv t + 1 \pmod{p}$. Thus (15) becomes

$$\eta_{t+1} = \theta\eta_t + \alpha U_{t+1} + (t + 1)\Xi U_t.$$

Insert this formula into the definitions of $\mathcal{C}_{d+1}$ and $\mathcal{M}_{d+1}$. Since the characteristic is p , $\theta(X^p) = 0$, and hence $\theta(X^p U_t) = X^p U_{t+1}$. Collecting coefficients gives (23).

Here $\Xi = C/12$ remains separate from $\theta\alpha$. In general the Serre derivative of E_{p-1} can contain a cusp-form term, so identifying these two divided series would not be valid uniformly. We use only Lemma 5.5, which gives $\theta\alpha + X^p = X$.

For the second recurrence, use in addition

$$\theta X = X^2 - \frac{E_4}{144}.$$

The product rule gives

$$\theta(X^d U_s) = dX^{d+1}U_s - \frac{d}{144}E_4X^{d-1}U_s + X^dU_{s+1}.$$

After multiplication by $(d-1)!$, the first term cancels the $d!X^{d+1}U_s$ term in $\mathcal{M}_{d+1}$. The remaining terms are exactly (24). $\square$

Lemma 5.7 (A modular divided series). *The series*

$$Y := \Xi + X^p$$

is the q -expansion of a modular form in $M_{p+1}(\mathbb{F}_p)$.

Proof. The Serre derivative

$$\partial A = \theta A - \frac{p-1}{12}E_2A$$

and B are modular forms of weight $p+1$. Since $A \equiv 1 \pmod{p}$, one has $12\partial A \equiv B \pmod{p}$. Hence

$$H := \frac{12\partial A - B}{12p} \pmod{p}$$

belongs to $M_{p+1}(\mathbb{F}_p)$ by the integral q -expansion principle. From $B - AE_2 = 12p\Xi$ and $A = 1 + p\alpha$ one obtains

$$H = \theta\alpha - X - \Xi.$$

Lemma 5.5 gives $H = -(\Xi + X^p) = -Y$, which proves the claim. $\square$

The recurrence errors are not new obstruction classes. They already lie below the next lower Hasse cutoff.

Corollary 5.8 (The recurrence errors vanish in the next quotient). *Fix m , and let $w_{m,d} = k + 2i - p(p-1)$ for the cell with parameter d . If*

$$1 \leq d \leq k - 2m - 2,$$

then the right sides of (23) and (24) belong to $M_{w_{m,d+1}}(\mathbb{F}_p)$. Consequently

$$\begin{aligned} [\mathcal{C}_{d+1}] &= [\theta\mathcal{C}_d], \\ [\mathcal{M}_{d+1}] &= [\theta\mathcal{M}_d] \end{aligned}$$

after reduction modulo $M_{w_{m,d+1}}(\mathbb{F}_p)$.

Proof. The terms YU_{s+d} and XU_{s+d} have filtration at most

$$F_1 = k + 2s + 2 + d(p+1).$$

The same bound holds for X^dU_{s+1} . Direct substitution gives

$$F_1 - w_{m,d+1} = -(p-2-d-2m)(p-1) < 0.$$

The remaining term $E_4X^{d-1}U_s$ has filtration

$$F_2 = k + (s+d-1)(p+1) + 4,$$

and

$$F_2 - w_{m,d+1} = (d-k+2m+1)(p-1) < 0.$$

Both strict inequalities follow from $d \leq k - 2m - 2$ and $k < p - 1$. Thus every recurrence-error term has a representative of weight at most $w_{m,d+1}$, as claimed. $\square$

Corollary 5.8 removes every inhomogeneous source from the proposed induction in d . It does not by itself finish that induction: applying θ to a form of weight $w_{m,d}$ naturally produces weight $w_{m,d} + p + 1$, whereas the next cutoff is $w_{m,d} + 2$. The remaining step is therefore one precise Hasse-descent claim for the differentiated previous class, not another factorial or binomial identity.

Lemma 5.9 (Unit-block endpoint coefficient). *Suppose that $i = np + i'$ lies on either of the two triangular lines in (5), and put $m = p - 2 - n$. Define*

$$\kappa_A = \frac{1}{p} \binom{i}{i-p} \prod_{r=i-p+k}^{i+k-1} r \pmod{p}.$$

Then

$$\kappa_A = -n(n+1) = -(m+1)(m+2) \pmod{p},$$

and the coefficient of the $X^p \theta^{i-p} f$ unit block after subtracting the shifted coefficient $q_1(n, i')$ is

$$\kappa_A + i(i+k-1) - q_1(n, i') = -2(m+1) \pmod{p}. \quad (25)$$

Proof. The quotient defining κ_A is an integer because the product contains a factor divisible by p . Lucas's theorem gives $\binom{i}{i-p} = \binom{i}{p} \equiv n \pmod{p}$. On the first triangular line one has $i' = p - k + m + 1$, and on the second one has $i' = p - k + m + 2$. In either case the product defining κ_A consists of p consecutive integers and its unique multiple of p is $(n+1)p$. After division by p , the remaining factors run through $\mathbb{F}_p^\times$. Wilson's theorem therefore gives

$$\frac{1}{p} \prod_{r=i-p+k}^{i+k-1} r \equiv -(n+1) \pmod{p},$$

which proves the formula for κ_A . Since $i(i+k-1) \equiv i'(i'+k-1)$ and

$$q_1(n, i') = i'(i'+k-1) - (n+1)(n+2),$$

their difference is

$$-n(n+1) + (n+1)(n+2) = 2(n+1) = -2(m+1) \pmod{p}.$$

□

Lemma 5.10 (The finite block has no triangular endpoint). *On either triangular line, the divided finite block obtained after the inversion identity has the form*

$$\mathcal{F}_i = \sum_{\ell=np}^{np+p-k} b_{i,\ell} X^{i-\ell} \theta^\ell f \quad (b_{i,\ell} \in \mathbb{F}_p). \quad (26)$$

On the line $i' + n = 2p - k - 1$, its class belongs to V_m^- . On the line $i' + n = 2p - k$, its class belongs to V_m^+ . Consequently its coefficient at the distinguished X^p endpoint is zero on both lines for every admissible p, k, m .

Proof. For completeness, first consider the lower exponents produced by the inversion. Put

$$a_{i,j} = \binom{i}{j} \frac{(i+k-1)!}{(j+k-1)!}, \quad b_{j,\ell} = (-1)^{j-\ell} \binom{j}{\ell} \frac{(j+k-1)!}{(\ell+k-1)!}.$$

For $\ell < np$, the coefficient K_ℓ after division by p satisfies

$$pK_\ell = a_{i,\ell} \sum_{u=np-\ell}^{np+p-k-\ell} (-1)^u \binom{i-\ell}{u}.$$

This follows by substituting the inversion formula and using $\binom{i}{j} \binom{j}{\ell} = \binom{i}{\ell} \binom{i-\ell}{j-\ell}$. To make the divisibility explicit, write $\ell = up + b$ with $0 \leq b < p$. Because $\ell < np$, one has $u \leq n-1$. Legendre's formula and Kummer's theorem give

$$v_p(a_{i,\ell}) = n + 1 - u - \mathbf{1}_{b \geq p-k+1} + \mathbf{1}_{b > i'}.$$

This valuation can equal 1 only when $u = n-1$ and $p-k+1 \leq b \leq i'$. This is exactly the interval $np-k+1 \leq \ell \leq i-p$. Thus $p^2 \nmid a_{i,\ell}$ outside that interval. If ℓ lies inside it, put $q = i-p-\ell$. Then $0 \leq q < p$,

$$i-\ell = p+q, \quad np-\ell > q, \quad np+p-k-\ell < p.$$

Lucas's theorem gives $\binom{p+q}{u} \equiv 0 \pmod{p}$ throughout the displayed summation range. In both cases $p^2 \nmid pK_\ell$, so $K_\ell \equiv 0 \pmod{p}$. This proves that the divided finite block has the range in (26).

Write $\ell = np+t$, where $0 \leq t \leq p-k$. Since $n = p-2-m$, coefficientwise Fermat periodicity gives

$$\theta^{np+t} f = \begin{cases} \theta^{p-2-m+t} f, & 0 \leq t \leq m+1, \\ \theta^{t-m-1} f, & m+2 \leq t \leq p-k, \end{cases} \quad \text{in } \mathbb{F}_p[[q]].$$

On the first triangular line, $i' = p-k+m+1$. For $0 \leq t \leq m+1$, the corresponding term in (26) is

$$X^{p-k+m+1-t} \theta^{p-2-m+t} f.$$

The two exponents add to $2p-k-1$, and the X exponent lies in the range defining V_m^- . For $t \geq m+2$, the two exponents add to $p-k$. Such a term is congruent to a modular form of weight

$$F_- = k + (p-k)(p+1).$$

A direct substitution gives

$$F_- - w_- = (2m+2-k)(p-1) \leq 0.$$

It therefore lies in $M_{w_-}(\mathbb{F}_p)$ and is zero in the quotient. This proves that $[\mathcal{F}_i] \in V_m^-$.

On the second triangular line, $i' = p-k+m+2$. For $0 \leq t \leq m+1$, the term is

$$X^{p-k+m+2-t} \theta^{p-2-m+t} f,$$

whose exponents add to $2p-k$ and whose X exponent lies in the range defining V_m^+ . For $t \geq m+2$, the exponents add to $p-k+1$, so the term has a modular representative of weight

$$F_+ = k + (p-k+1)(p+1).$$

Here

$$F_+ - w_- = (2m+3-k)(p-1) \leq 0.$$

Thus these terms also vanish in the lower quotient, and $[\mathcal{F}_i] \in V_m^+$. Proposition 5.12 shows that the omitted endpoint is independent of each preceding span. Its coefficient in $\mathcal{F}_i$ is therefore zero. $\square$

Lemma 5.11 (Exact filtration after multiplication by X). *Let g be a nonzero mod- p modular form with exact filtration $\omega_p(g) = v$. For every $a \geq 0$,*

$$\omega_p(X^a g) = v + a(p + 1).$$

Here X is viewed as the mod- p modular form $E_{p+1}/12 \equiv E_2/12$.

Proof. Choose a representative of g in $M_v(\mathbb{F}_p)$. The standard Hasse-divisibility criterion says that its weight is minimal exactly when it is not divisible by the Hasse invariant A [11, 8]. By Lemma 4.8, X is a unit along the Hasse divisor. Multiplication by X^a therefore does not change the order of vanishing along that divisor, so $X^a g$ is still not divisible by A . It has weight $v + a(p + 1)$. Applying the same Hasse-divisibility criterion once more shows that this weight is minimal. $\square$

Proposition 5.12 (Uniform separation of the triangular endpoint). *Assume that f is nonzero modulo p , is ordinary at p , has $\omega_p(f) = k$, that $4 \leq k < p - 1$, and that a $D = 0 - D = 2$ cross-wedge cell lies on one of the two triangular lines in (5). Put $m = p - 2 - n$. Then*

$$1 \leq m \leq \frac{k - 4}{2}.$$

On the line $i' + n = 2p - k - 1$, the class

$$X^p \theta^{p-k-1} f$$

is not in $V_m^- + M_{w-}(\mathbb{F}_p)$. On the line $i' + n = 2p - k$, the class

$$X^p \theta^{p-k} f$$

is not in $V_m^+ + M_{w-}(\mathbb{F}_p)$. In particular, the endpoint classes in (9) and (10) are independent of the preceding triangular spans for every admissible p, k, m , not only in the finite audit.

Proof. Let $\ell_0 = p - k + 1$ be the first low point of the ordinary mod- p theta cycle. For $0 \leq t < p$, its filtration is

$$\omega_p(\theta^t f) = \begin{cases} k + t(p + 1), & t < \ell_0, \\ k + t(p + 1) - (p - k + 1)(p - 1), & t \geq \ell_0. \end{cases}$$

Lemma 5.11 gives

$$\omega_p(X^a \theta^t f) = \omega_p(\theta^t f) + a(p + 1).$$

On the first triangular line, every generator of V_m^- has $a + t = 2p - k - 1$ and

$$p - 2m - 2 \leq t \leq p - 1.$$

The cross-wedge inequalities give $2m + 2 \leq k - 2$, hence $t \geq p - k + 2 > \ell_0$. All these generators therefore have exact filtration

$$F_-^{\text{old}} = k + (2p - k - 1)(p + 1) - (p - k + 1)(p - 1) = p^2 + p - k.$$

The endpoint has $a = p$ and $t = p - k - 1 < \ell_0$, so its exact filtration is

$$F_-^{\text{end}} = k + (2p - k - 1)(p + 1) = p(2p + 1 - k) - 1.$$

Thus

$$F_-^{\text{end}} - F_-^{\text{old}} = (p - k + 1)(p - 1) > 0.$$

Moreover a direct substitution in $w_- = k + 2i - p(p - 1)$ gives

$$F_-^{\text{old}} - w_- = 2(m + 1)(p - 1) > 0.$$

On the second triangular line, every generator of V_m^+ has $a + t = 2p - k$ and

$$p - 2m - 3 \leq t \leq p - 1.$$

Here $2m + 3 \leq k - 1$, so again $t \geq \ell_0$. Their common exact filtration is

$$F_+^{\text{old}} = k + (2p - k)(p + 1) - (p - k + 1)(p - 1) = p^2 + 2p - k + 1.$$

The endpoint has $a = p$ and $t = p - k < \ell_0$, and hence

$$F_+^{\text{end}} = k + (2p - k)(p + 1) = p(2p + 2 - k).$$

Again

$$F_+^{\text{end}} - F_+^{\text{old}} = (p - k + 1)(p - 1) > 0, \quad F_+^{\text{old}} - w_- = (2m + 3)(p - 1) > 0.$$

If an endpoint belonged to the sum of the preceding span and M_{w_-} , its filtration would be at most the larger of w_- and the filtrations of the preceding generators. The strict inequalities above contradict the exact endpoint filtration. This proves both assertions. $\square$

Lemma 5.10 proves that the triangular B -block sum has zero distinguished-endpoint coefficient on both lines. The exact audit further shows that the explicit first-Hasse-jet correction in (13) has coefficient $m + 1$ in all 156 tested cells. Together with (25), this gives

$$-2(m + 1) + 0 + (m + 1) = -(m + 1).$$

Thus the open uniform statement on these two lines is now the associated-graded simplification

$$\Pi_i \equiv (m + 1)X^p\theta^{p-k-\varepsilon}f \pmod{V_m^\varepsilon}, \quad \varepsilon = \begin{cases} 1, & i' + n = 2p - k - 1, \\ 0, & i' + n = 2p - k, \end{cases} \quad (27)$$

where $V_m^1 = V_m^-$ and $V_m^0 = V_m^+$. The independence of the displayed endpoint is Proposition 5.12; the first-Hasse-jet formula itself is Theorem 5.2. Only the uniform triangular containment of the remaining core series is open. The unit block, finite block, and low-point jet corrections are now proved.

The two open endpoint simplifications have a common core. The next proposition records it exactly. The low-point proposition then removes the jet corrections and leaves one direct block-basis containment.

Proposition 5.13 (A common core for the two endpoint identities). *Put $s = p - k$, $U_t = \theta^t f$, and define*

$$\begin{aligned} T_s &= -\Phi_s + 2\alpha U_s + (k + 1)X^p U_{s-1}, \\ T_{s+1} &= -\Phi_{s+1} + 2\alpha U_{s+1} + (k - 1)X^p U_s, \\ S_s &= \alpha U_s + X^p U_{s-1}. \end{aligned}$$

On the first triangular line, the endpoint identity is equivalent to

$$-\eta_s + (m + 1)T_s - kS_s \in V_m^- + M_{w_-}(\mathbb{F}_p). \quad (28)$$

On the second triangular line, it is equivalent to

$$-\eta_{s+1} + (m + 1)T_{s+1} \in V_m^+ + M_{w_-}(\mathbb{F}_p). \quad (29)$$

Moreover,

$$T_{s+1} = \theta T_s - 2XU_s. \quad (30)$$

The term XU_s belongs to $M_{w_-}(\mathbb{F}_p)$ on the second line. Thus, after the lower cutoff, the same core series T_s controls both endpoint problems. Proposition 5.3 below the displayed cutoffs removes the two jet terms. The only remaining endpoint problem is the triangular containment of T_s and its theta image.

Proof. On the first line one has $d = 0$, $L = -(m+1)$, $a = k - 2m - 2$, and $q_1 = -km$. Substitute these values in (13), subtract $\mathcal{L}_i$, and add the target endpoint $(m+1)X^pU_{s-1}$. The result is exactly the left side of (28).

On the second line, put $d = 1$ in Proposition 5.4. Since $Z_1 = XU_s$ and

$$\omega_p(Z_1) - w_- = (2m + 3 - k)(p - 1) < 0,$$

the term mZ_1 lies below the cutoff. Adding the target endpoint $(m+1)X^pU_s$ then gives (29).

Finally, $\Phi_{s+1} = \theta\Phi_s$, $\theta(X^p) = 0$, and Lemma 5.5 gives $\theta\alpha = X - X^p$. Applying the product rule to T_s yields

$$\theta T_s = -\Phi_{s+1} + 2\alpha U_{s+1} + 2XU_s + (k-1)X^pU_s,$$

which is (30). The filtration statement for XU_s is the displayed gap above. $\square$

Corollary 5.14 (Exact remaining endpoint target). *On the first triangular line, the endpoint identity is equivalent to*

$$T_s \in V_m^- + M_{w_-}(\mathbb{F}_p).$$

On the second triangular line, it is equivalent to

$$T_{s+1} \in V_m^+ + M_{w_-}(\mathbb{F}_p).$$

The two series satisfy $T_{s+1} = \theta T_s - 2XU_s$, and the last term is below the second-line cutoff. Thus the open endpoint calculation is a triangular block-basis containment for one core series and its theta image. It is no longer a comparison of the pre-low-point and reset jets.

Proof. For the first line, rewrite (28) as

$$(m+1)T_s - (\eta_s + kS_s).$$

The second term lies below the cutoff by Proposition 5.3. For the second line, η_{s+1} lies below the cutoff by the same proposition. Since $1 \leq m < p-1$, the scalar $m+1$ is a unit in $\mathbb{F}_p$. The two equivalences follow, and the last statement is (30). $\square$

Figure 2 summarizes the reduction. It also marks the precise point where the proof stops: the two green jet terms are theorems, while the two dashed triangular containments remain open.

There is a parallel uniform statement on the third line. It does not determine the scalar in (6), but it proves that the displayed block is a genuine, nonzero quotient direction.

Proposition 5.15 (The third-line block is nonzero). *Assume the hypotheses of Proposition 5.12 and suppose that $i' = n+2$. Put $m = p-2-n$. Then $i' = p-m$, $1 \leq m \leq (k-4)/2$, and*

$$Y_{p,k,m} := X^{k-2m-1}\theta^{p-k}f$$

has exact filtration

$$\omega_p(Y_{p,k,m}) = w_- + (p-1).$$

In particular, its image in the lower Hasse quotient is nonzero. Also

$$q_1(n, i') = -km \not\equiv 0 \pmod{p}.$$

Proof. The equality $i' = n+2$ and the definition $m = p-2-n$ give $i' = p-m$. The cross-wedge inequalities give $1 \leq m \leq (k-4)/2$. Since $p-k$ lies strictly before the first low point $p-k+1$ of the ordinary theta cycle,

$$\omega_p(\theta^{p-k}f) = k + (p-k)(p+1).$$

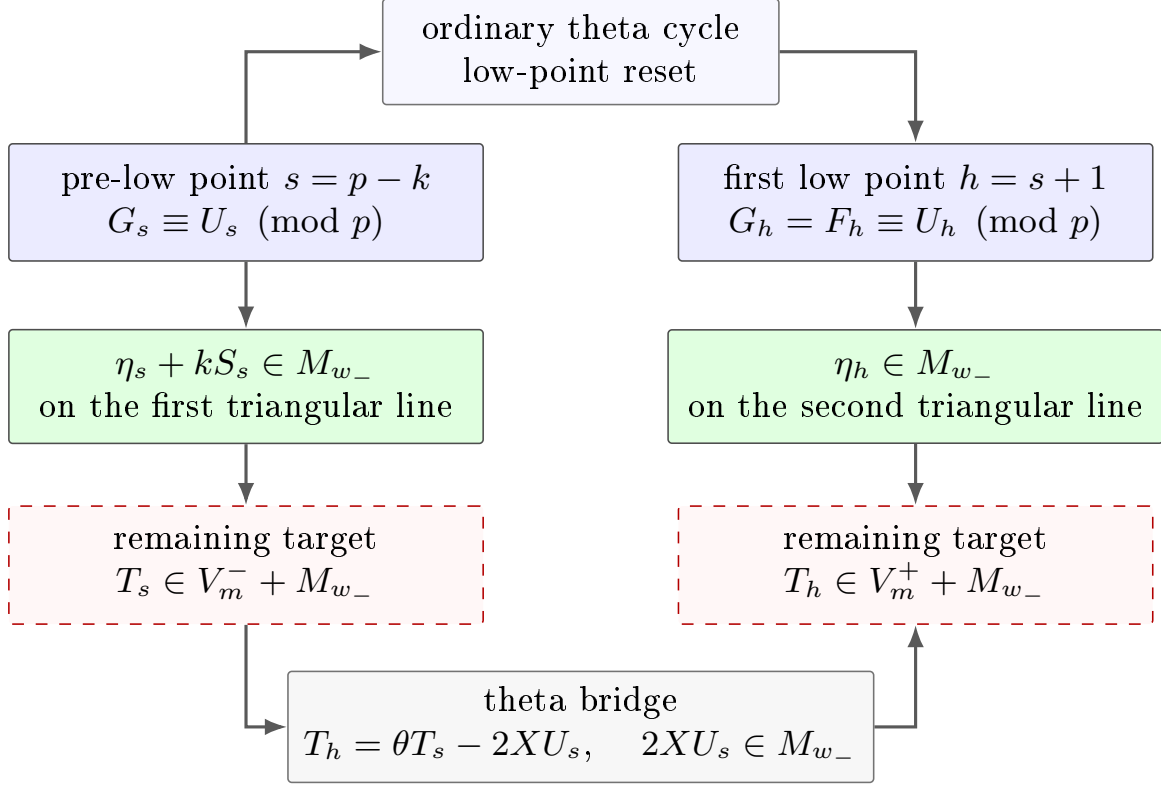


Figure 2: The first low point and the two endpoint reductions. The green boxes are proved lower-weight statements. After removing these jet terms, the two endpoint identities reduce to the dashed triangular containments. The theta bridge links their remaining core series.

Multiplication by X^{k-2m-1} raises exact filtration by $(k-2m-1)(p+1)$. Hence

$$\omega_p(Y_{p,k,m}) = k + (p-2m-1)(p+1).$$

On this line

$$i = (p-2-m)p + (p-m) = p^2 - p - m(p+1),$$

so

$$w_- = k + 2i - p(p-1) = k + p^2 - p - 2m(p+1).$$

Subtracting gives $\omega_p(Y_{p,k,m}) - w_- = p-1$. Thus $Y_{p,k,m}$ cannot lie in $M_{w_-}(\mathbb{F}_p)$ and its quotient class is nonzero. Finally, direct substitution of $n = p-2-m$ and $i' = p-m$ in q_1 gives $q_1 = -km$. Both k and m lie strictly between 0 and p , so this scalar is nonzero. $\square$

If the residual belongs to the line spanned by $Y_{p,k,m}$, its scalar in (6) is uniquely determined. Under this one-line hypothesis it can be read in canonical quotient coordinates as

$$c_{p,k,m}(f) = \frac{\lambda(\mathcal{B}_i - \mathcal{L}_i)}{\lambda(Y_{p,k,m})}, \quad (31)$$

where λ is any linear functional nonzero on the one-dimensional line spanned by $Y_{p,k,m}$. Because the residual is then a scalar multiple of $Y_{p,k,m}$, the ratio is independent of λ . Exact reconstruction of the computed values identifies the following closed candidate:

$$c_{p,k,m}^* = -m(k-2m-2)! \pmod{p}. \quad (32)$$

Proposition 5.16 (The closed third-line scalar is a unit). *For every admissible p, k, m , the value in (32) belongs to $\mathbb{F}_p^\times$.*

Proof. The cross-wedge inequalities give

$$1 \leq m \leq \frac{k-4}{2}, \quad 2 \leq k-2m-2 \leq k-4 < p.$$

Thus neither m nor any factor in $(k-2m-2)!$ is divisible by p . Their product is nonzero modulo p . $\square$

The finite evidence for this stronger statement is broader than the discovery table. A blind extension through $p = 79$ checked 652 third-line cells and found no zero scalar, no failure of the one-block shape. A separate weight-24 test, where the cusp-form space is two-dimensional, checked 1,860 cells from 186 ordinary forms and linear combinations through $p = 43$. In both audits every computed scalar was exactly (32); there were no mismatches. The scalar was also independent of the form for fixed p, k, m . These exact computations identify the formula and strongly test it; they do not replace the symbolic block reduction below.

The generic identity and the third-line formula now have one common target. For the parameters of Theorem 5.2, put

$$Z_d = X^d \theta^{p-k} f.$$

A direct filtration calculation gives

$$\omega_p(Z_d) - w_- = (d - k + 2m + 2)(p - 1). \quad (33)$$

Consequently $Z_d \in M_{w_-}(\mathbb{F}_p)$ for $d \leq k - 2m - 2$, whereas $Z_{k-2m-1} = Y_{p,k,m}$ is the nonzero direction of Proposition 5.15. It is therefore enough to prove the following single symbolic reduction.

Conjecture 5.17 (Factorial block reduction). *For $2 \leq d \leq k-2m-1$, the explicit first-Hasse-jet class satisfies*

$$\mathcal{B}_i - \mathcal{L}_i = -m(d-1)![Z_d] \quad \text{in } Q_{k+2i}. \quad (34)$$

For $2 \leq d \leq k - 2m - 2$, equation (33) makes the right side of (34) zero, proving the generic shifted identity. At $d = k - 2m - 1$, the same equation becomes (6) with the explicit scalar (32), which is a unit by Proposition 5.16. Thus one factorial reduction proves both the generic identity and the complete third-line assertion.

Theorem 5.2 replaces the formerly unspecified minimal-weight projection compatibility square by an exact operator formula. What remains is the formal reduction of (13) to (34), together with the two triangular core-series containments in Corollary 5.14. The audit identifies the resulting endpoint coefficient as the unit $-(m+1)$. Proposition 5.12 already supplies endpoint independence; it must not be counted among the remaining conjectural inputs.

6 Interpretation of the experiments

The main experiment finds one simple pattern. One quadratic controls the interior of the cross-wedge. Three affine boundary lines have extra quotient directions. The same failures remain when we raise the Fourier cutoff. They are separate filtration cases, not truncation errors.

The proofs explain the following parts of this pattern for all admissible parameters. The first-Hasse-jet formula explains why the $D = 2$ block fails in the new wedge. Exact filtration shows that the three boundary directions remain nonzero in the quotient. The perfect Hasse-quotient pairing and theta skew-adjointness connect the two obstructions. These theorems do not give every boundary coefficient. The missing steps are the factorial block reduction in Conjecture 5.17 and two triangular containments for the common endpoint core. We now compute the pre-low-point and reset jet terms for all admissible parameters. Both lie below the relevant cutoffs.

The evidence is strongest on the third boundary line. We first observed the one-block shape in 78 cells. A separate blind audit tested 652 cells. A second audit used 186 ordinary forms

in a two-dimensional cusp-form space. Together, the two later audits made 2,512 tests. Every test returned the same form-independent factorial scalar. This is strong finite evidence for the stated conjecture. It does not replace the missing symbolic proof.

The study has three limits. First, the main cross-wedge table ends at $p = 43$. The separate third-line test reaches $p = 79$, and the symbolic index audit reaches $p = 101$. Second, the broad record uses level-one ordinary forms. It does not test supersingular or higher-level behavior. Third, the two triangular coefficient identities have exact finite support, proved endpoint independence, and proved low-point jet reductions. The uniform containment of the remaining core series is still open. These limits define the next tests and proof problems.

7 Tate-dual obstruction classes

Put $P = p(p - 1)$ and let $i = np + i'$ be in the nondegenerate band. Define

$$i^\dagger = p^2 + 1 - k - i = (p - 2 - n)p + (2p + 1 - k - i').$$

On every admissible cell considered here, the mod- p^2 filtration has the same weight class as $k + 2i$. We therefore write

$$\omega_{p^2}(\theta^i f) = k + 2i + \delta(\theta^i f)P.$$

This equation defines the integer $\delta(\theta^i f)$. Thus $\delta = -1$ means that one additional full Hasse drop occurs below $k + 2i$, while $\delta = 0$ means that it does not. Then

$$i + i^\dagger = p^2 + 1 - k, \quad (k + 2i) + (k + 2i^\dagger) = 2p^2 + 2.$$

Moreover, if $c(x) = x^2 + (k - 1)x$, then $c(2p + 1 - k - i') \equiv c(i') \pmod{p}$, while

$$(p - 2 - n)(p - 1 - n) \equiv (n + 1)(n + 2) \pmod{p}.$$

The lower Bockstein class $\beta_w(g)$ and its lift-independence are given by Lemma 4.2. In particular,

$$\beta_w(g) = 0 \iff \omega_{p^2}(g) \leq w$$

for an admissible candidate weight w .

The perfect Hasse-quotient pairing and theta adjointness are proved in Propositions 4.5 and 4.7. Put

$$w_- = k + 2i - P, \quad w_+ = k + 2i^\dagger + P.$$

Proposition 4.10 places the two actual weight-lowering obstructions in

$$Q_{r_-} \quad \text{and} \quad Q_{r_+}, \quad r_- = w_- + P = k + 2i, \quad r_+ = w_+ + P = k + 2i^\dagger + 2P.$$

Their weights satisfy

$$r_- + r_+ = P + 2 + p(3p - 1).$$

Theorem 4.13 uses a horizontal unit of weight $p(3p - 1)$ to give a perfect pairing

$$[\ , \]^{\text{nat}} : Q_{r_-} \times Q_{r_+} \longrightarrow \mathbb{F}_p$$

as part of a correctly typed family under which theta is skew-adjoint. The pairing shows that the two obstruction spaces are dual. It does not by itself identify the two particular obstruction classes.

Conjecture 7.1 (Cross-wedge Bockstein compatibility). *Assume that f is nonzero modulo p , is ordinary at p , has $\omega_p(f) = k$, $4 \leq k < p - 1$, and that both i and $i^\dagger$ lie in the nondegenerate band. If*

$$D(i) = 0, \quad D(i^\dagger) = 2,$$

then the two geometric Bockstein classes

$$\Gamma_{w_-}(\theta^i f) \in Q_{r_-}, \quad \Gamma_{w_+}(\theta^{i^\dagger} f) \in Q_{r_+}$$

vanish together:

$$\Gamma_{w_-}(\theta^i f) = 0 \iff \Gamma_{w_+}(\theta^{i^\dagger} f) = 0.$$

Equivalently,

$$\beta_{k+2i-P}(\theta^i f) = 0 \iff \beta_{k+2i^\dagger+P}(\theta^{i^\dagger} f) = 0.$$

The conjecture states only the intrinsic vanishing equivalence needed below. A stronger equality through a chosen auxiliary-level pairing would require an explicit comparison map and a normalization of the horizontal unit. Those data have not yet been constructed, so no such equality is claimed here.

Conjecture 7.2 (Exact $D = 0$ shifted-conic law). *Under the hypotheses of the preceding conjecture,*

$$\delta(\theta^i f) = -1 \iff i'^2 + (k-1)i' - (n+1)(n+2) \equiv 0 \pmod{p}.$$

Otherwise $\delta(\theta^i f) = 0$.

Proposition 7.3 (Conditional shifted-conic deduction). *Assume Conjecture 7.1. Assume also that the complementary $D = 2$ cell lies in the exact range where*

$$\Gamma_{w_+}(\theta^{i^\dagger} f) = 0 \iff (i'^\dagger)^2 + (k-1)i'^\dagger - n^\dagger(n^\dagger + 1) \equiv 0 \pmod{p}.$$

Then the vanishing equivalence in Conjecture 7.2 holds.

Proof. For the dual index, the proved $D = 2$ polynomial is

$$(i'^\dagger)^2 + (k-1)i'^\dagger - n^\dagger(n^\dagger + 1).$$

The two elementary congruences above transform this into

$$i'^2 + (k-1)i' - (n+1)(n+2).$$

The additional $D = 2$ hypothesis says that the complementary obstruction vanishes exactly when this polynomial is zero. The assumed cross-wedge compatibility transfers this vanishing to i . This gives the claimed equivalence. Ahlgren–Raum–Richter [2, Theorem C] motivate this hypothesis: they prove exact values off the exceptional locus in the stated ranges and a sharper upper bound on that locus. Their theorem alone is not being used here as an exact vanishing equivalence in our coordinates. $\square$

Conjecture 7.4 (Corrected global duality). *Let $e(i) = D(i) - \delta(\theta^i f)$. Away from the central $D = 1$ plateau,*

$$e(i) = e(i^\dagger).$$

On that plateau the difference is a rank-one boundary functional supported on

$$i'^2 + (k-1)i' - (n+1)^2 \equiv 0 \pmod{p}.$$

8 Remaining symbolic identities

Theorem 5.2 constructs the minimal mod- p representative in characteristic zero and computes its normal coefficient along the Hasse divisor. A complete cross-wedge theorem now requires the factorial block reduction in Conjecture 5.17 and the two triangular associated-graded containments in Corollary 5.14. The residual directions, their independence, the finite-block endpoint contribution, and both low-point jet corrections are already proved. Proposition 5.13 links the two remaining endpoint formulas by one exact theta recurrence. The open task is to place the core series T_s and its theta image in the stated triangular spans, and to simplify the factorial operator formula in the direct block basis.

On the third line, that simplification predicts

$$\mathcal{B}_i - \mathcal{L}_i = -m(k - 2m - 2)!X^{k-2m-1}\theta^{p-k}f.$$

The displayed block is nonzero by Proposition 5.15, and the scalar is a unit by Proposition 5.16. Thus the unresolved point is not nonvanishing, but identification of the computed scalar with the formal block reduction for every admissible parameter set.

9 Conclusion

The exact experiments find a shifted quadratic inside the next theta-cycle wedge and three extra residuals on its boundary. The geometry and filtration results put these residuals in the predicted quotient directions. The first Hasse jet and the direct block calculation reduce the open problem to three coefficient identities. The finite block has no triangular endpoint. One exact theta recurrence links the two remaining endpoint cores. Closed formulas remove the pre-low-point and reset jets from the open part. The factorial block reduction and the two triangular containments remain open. The table below states which results are uniform theorems, finite results, conditional claims, or conjectures.

10 Theorem status

The table states the status of each claim used in the shifted-conic application.

Statement	Status	Scope or limitation
Bockstein realization as a first Hasse jet; perfect Hasse-quotient pairing; theta skew-adjointness	Proved uniformly	Geometric results on the admissible level cover, with descent and indexing made explicit.
Integral Hasse inclusion and explicit periodic first-Hasse-jet formula	Proved uniformly	Ordinary level-one forms with $4 \leq k < p - 1$ in the admissible cross-wedge.
Boundary-direction separation and nonvanishing of the factorial scalar candidate	Proved uniformly	Establishes the quotient directions and that $-m(k - 2m - 2)!$ is a unit; it does not identify the actual residual scalar.
Affine factorial reduction, its two exact recurrences, and exclusion of the finite block from both triangular endpoints	Proved uniformly	Reduces the factorial target to explicit divided series. The endpoint targets have one common core linked by an exact theta recurrence. Closed low-point formulas put both jet corrections below the endpoint cutoffs.
Direct obstruction pattern and exact scalar agreement	Verified exactly in finite range	692 cross-wedge cells and 2,512 third-line tests, all Sturm-safe and exactly reproducible.
Factorial block reduction and two triangular containments for the endpoint core series	Conjectural	The remaining direct block-basis calculation; the low-point jet comparison is no longer open.
Cross-wedge Bockstein compatibility, shifted-conic law, and corrected global duality	Conditional	Follow from the stated block and compatibility conjectures; they are not used to prove the unconditional theorems.

A Computational reproducibility

A.1 Scope and arithmetic

The code has three jobs. It rejects false formulas, finds exact boundary patterns, and checks that the explicit constructions give the canonical remainders. A finite check is never used as a proof for all primes. Every modular reduction and linear system is evaluated over $\mathbb{Z}/p^2\mathbb{Z}$ or $\mathbb{F}_p$ with integer arithmetic. The reported audits use no floating-point tolerance, random seed, or random search.

The reference environment is Python 3.13.7 with NumPy 2.4.6 and SymPy 1.14.0. From the project directory it can be installed with

```
python -m pip install -r requirements.txt
```

The deposited comma-separated files in `data/` contain the cell parameters and canonical obstruction coordinates. The implementation in `code/branchfree.py` constructs level-one echelon bases and keeps enough Fourier coefficients to pass the Sturm bound at every tested weight.

A.2 Primary checks

All seven main audits can be run with

```
python run_core_audit.py
```

The script stops with a nonzero exit code if a command fails or if an expected exact summary is missing. The individual commands are given below.

The explicit lower-weight lift and periodic first-Hasse-jet formula are checked by

```
python verify_periodic_first_jet.py data/d0_obstruction_coordinates.csv
```

The expected terminal summary is

```
rows=692 checked=692 bad_weight=0 bad_mod_p=0 bad_beta=0
```

Here `bad_weight` tests the exact weight of the constructed lift, `bad_mod_p` tests its congruence with $\theta^i f$, and `bad_beta` compares the resulting divided series with the canonical echelon Bockstein.

The blind third-line extension is

```
python stress_test_l3_uniformity.py --max-p 79
```

and ends with

```
checked=652 zero=0 failed_shape=0 wrong_scalar=0
```

The independent weight-24 test is

```
python stress_test_l3_weight24.py --max-p 43
```

and ends with

```
ordinary_forms=186 checked=1860 zero=0 failed_shape=0 wrong_scalar=0
```

In both commands, `wrong_scalar` compares the computed value with $-m(k - 2m - 2)!$.

The symbolic index audit

```
python audit_index_formulas.py
```

checks the exponent decompositions, lower-weight formulas, Hasse exponents, pairing sums, and filtration gaps over all admissible symbolic parameters for primes through 101. This audit is deliberately separate from the Fourier-coefficient calculation.

The affine reduction and its two recurrences are checked by

```
python audit_factorial_affine_reduction.py
```

The expected summary is

```
affine_checked=614 recurrence_checked=156 failures=0
```

The endpoint source decomposition is checked by

```
python audit_triangular_endpoint_source.py \
  data/d0_obstruction_coordinates.csv --examples 0
```

It reports 78 matches on each line for the unit block, zero finite-block endpoint, and first-Hasse-jet correction. A failure category is never suppressed.

The low-point jet proof is audited independently by

```
python audit_low_point_jets.py
```

The expected summary is

```
families=15 triangular_identities=273 modularized_jets=273
first_cutoffs=78 second_cutoffs=78 failures=0
```

This command checks both triangular modified-Serre identities, the modularized first-jet formula, the closed reset-jet formula, and all 156 cutoff containments.

A.3 Interpretation

The checks show that the programs implement the stated formulas on the deposited range. They also show that the first-Hasse-jet lift agrees with an independent canonical reduction. They do not prove Conjecture 5.17 or the two triangular core-series containments for arbitrary p . Those statements remain clearly separated from the unconditional Hasse-jet, low-point, and boundary-separation theorems.

Data and code availability

The public release contains the cell tables, obstruction coordinates, source code, and commands used for every computational count in this paper. It contains only research files. Code, data, and versioned releases are in the public repository:

<https://github.com/jopoku16/theta-p2-hasse-jet-experiments>.

References

- [1] S. Ahlgren, M. Hanson, M. Raum, and O. K. Richter, *Eisenstein series modulo p^2* , Forum Math. **38** (2026), no. 1, 93–102, doi:10.1515/forum-2024-0470.
- [2] S. Ahlgren, M. Raum, and O. K. Richter, *Theta cycles of modular forms modulo p^2* , Canad. J. Math. (2026), First View, 1–18, published online 13 April 2026, doi:10.4153/S0008414X2610220X.
- [3] I. Chen and I. Kiming, *On the theta operator for modular forms modulo prime powers*, Mathematika **62** (2016), 321–336, doi:10.1112/S0025579315000212.
- [4] M. Chida and M. Kaneko, *On ordinary primes for modular forms and the theta operator*, Proc. Amer. Math. Soc. **135** (2007), 1001–1005, doi:10.1090/S0002-9939-06-08561-3.
- [5] R. F. Coleman and J. F. Voloch, *Companion forms and Kodaira–Spencer theory*, Invent. Math. **110** (1992), 263–281.
- [6] B. Edixhoven, *The weight in Serre’s conjectures on modular forms*, Invent. Math. **109** (1992), 563–594, doi:10.1007/BF01232041.
- [7] B. H. Gross, *A tameness criterion for Galois representations associated to modular forms (mod p)*, Duke Math. J. **61** (1990), 445–517.
- [8] N. M. Katz, *p -adic properties of modular schemes and modular forms*, in *Modular Functions of One Variable III*, Lecture Notes in Math. 350, Springer, 1973, 69–190.
- [9] N. M. Katz, *A result on modular forms in characteristic p* , in *Modular Functions of One Variable V*, Lecture Notes in Math. 601, Springer, 1977, 53–61.
- [10] J.-P. Serre, *Formes modulaires et fonctions zêta p -adiques*, in *Modular Functions of One Variable III*, Lecture Notes in Math. 350, Springer, 1973, 191–268.
- [11] H. P. F. Swinnerton-Dyer, *On ℓ -adic representations and congruences for coefficients of modular forms*, in *Modular Functions of One Variable III*, Lecture Notes in Math. 350, Springer, 1973, 1–55.

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